

The Vector Substrate: Number Fields as Exact Learning Geometry

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Abstract

We develop a framework in which a number field $K = \mathbb{Q}(\theta)$ is treated *simultaneously* as a $\mathbb{Q}$ -vector space, as a commutative matrix algebra (its regular representation, with multiplication by θ realised by the companion matrix of the minimal polynomial), and as a Euclidean lattice (its Minkowski embedding). The trace form $T(x, y) = \text{Tr}_{K/\mathbb{Q}}(xy)$ supplies a single, exact, rational metric G that unifies these three faces. We build an exact, fully reproducible *learning* apparatus on this geometry: an observation is captured precisely when its G -orthogonal residual against a forced basis vanishes, and learning is the gated, witnessed growth of that basis by adjoining algebraic-integer seeds and, when required, by passing to a disjoint compositum (whose Gram matrix is the Kronecker product of the factor Grams). Model complexity is controlled by arithmetic height: the Mahler measure $M(\theta) = H(\theta)^{\deg \theta}$, Landau's inequality, and Northcott finiteness yield an exact-integer admissibility gate. Our information-geometric reading *assembles classical facts*: the trace form is the Gram of the Minkowski embedding ($G = M^T M$, $\det G = d_K$), and the Fisher information of a Gaussian location family $\mathcal{N}(Ma, I)$ is $M^T M$. From these we read the trace form as the Fisher metric of two natural models, pin the model-dependent conformal constant ($c = 1$ for the Gaussian, $c = n$ for the maximum-entropy family on the Galois conjugates), and obtain a residual-norm-as-information-distance reading for the learner: on the trace-zero (residual) subspace $\|\mathbf{r}\|_G^2 = n \cdot \mathcal{I}(\mathbf{r})$. We do not claim to settle an open problem—the contribution is this synthesis and its use. We then derive an information gain-versus-cost growth threshold whose floor is provably nonzero for non-reciprocal seeds (Smyth's theorem) and discuss the reciprocal case (Lehmer's conjecture). Two further bridges tighten the picture: the regular representation is diagonalised by the Minkowski embedding, so its spectrum *is* the set of Galois conjugates and the Mahler measure is a spectral invariant of the companion matrix; and the Jeffreys volume of the Fisher metric equals $\sqrt{\det G} = \sqrt{|d_K|}$ (in an integral basis), identifying the field discriminant with an information volume. Throughout we separate *provable* statements (with proofs or exact computations) from *conjectural* or model-dependent ones; every numerical claim is matched to a machine-checked verification probe, and a notation key and an honesty ledger make the provable/conjectural split scannable.

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1 Introduction and thesis

The thesis

Let $K = \mathbb{Q}(\theta)$ be a number field of degree $n = [K : \mathbb{Q}]$, and let $m_\theta \in \mathbb{Z}[x]$ be the minimal polynomial of θ . The field K admits three classical descriptions:

- (i) a $\mathbb{Q}$ -vector space of dimension n , with basis $\{1, \theta, \dots, \theta^{n-1}\}$;
- (ii) a commutative subalgebra of $M_n(\mathbb{Q})$, via the *regular representation* $\rho: K \rightarrow M_n(\mathbb{Q})$, $x \mapsto$ (multiplication by x), under which $\rho(\theta)$ is the companion matrix $C(m_\theta)$;
- (iii) a full-rank Euclidean lattice in $\mathbb{R}^n$, via the Minkowski embedding $\sigma = (\sigma_1, \dots, \sigma_n)$ of the n archimedean places.

These are not three analogies but *one object viewed in three coordinate systems*, glued by a single rational bilinear form: the trace form

$$T(x, y) = \text{Tr}_{K/\mathbb{Q}}(xy) = \sum_{i=1}^n \sigma_i(x) \sigma_i(y). \quad (1)$$

The matrix $G = (\text{Tr}(\omega_i \omega_j))_{ij}$ of T in a basis $\{\omega_i\}$ is the bridge: it is the Gram matrix of the algebra's pairing, and simultaneously the Gram matrix of the embedded lattice.

Why this matters for learning

Equation (1) makes K a metric space in which membership is decidable by exact rational arithmetic. Fix a “forced” subspace spanned by a small dictionary of algebraic-integer seeds. An observation $\mathbf{x} \in K$ is *captured* exactly when its G -orthogonal residual $\mathbf{r} = \mathbf{x} - P\mathbf{x}$ is zero, where P is the trace-form projector onto the dictionary. Capture is decided over $\mathbb{Q}$; only its *magnitude* $\|\mathbf{r}\|_G$ is ever read as a real number. “Understanding” is, in this framework, exactly a zero residual; *learning* is the disciplined enlargement of the dictionary so as to capture persistent residuals.

Contributions

- (C1) **An exact trace-form learning geometry** (§3–4): the projector/residual calculus over $\mathbb{Q}$, a propose-for-confirm growth protocol with a tamper-evident witness chain, and an exact *coordinates-to-minimal-polynomial* bridge computed as the largest invariant factor of $xI - \rho(\alpha)$ via Smith normal form.
- (C2) **Cross-field growth** (§4.4): when a residual lies outside the current field, the model grows to a disjoint compositum, whose trace-form Gram is the Kronecker product $G_{KL} = G_K \otimes G_L$ (Proposition 4.5); the non-disjoint case is isolated and flagged open.
- (C3) **Height-based capacity control** (§5): an exact-integer admissibility gate founded on $M(\theta) = H(\theta)^{\deg \theta}$, Landau’s inequality, and Northcott finiteness, with a provable separation between the exact decision boundary and float-valued display quantities.
- (C4) **Information geometry (a synthesis)** (§7): the classical facts $G = M^\top M$ and $\mathcal{I}_{\text{Gauss}} = M^\top M$ make the trace form a Fisher metric; we pin the conformal constant ($c = 1$ Gaussian vs $c = n$ max-entropy) and show $\|\mathbf{r}\|_G^2 = n \mathcal{I}(\mathbf{r})$ on the residual subspace (Theorems 7.1–7.9).

- (C5) **Two structural bridges** (§2.2, §7): the regular representation is diagonalised by the Minkowski embedding, so its spectrum is the set of Galois conjugates and the Mahler measure is a spectral invariant of the companion matrix (Proposition 2.3, Remark 2.4); and the Jeffreys volume of the Fisher metric is $\sqrt{\det G} = \sqrt{|d_K|}$ (in an integral basis), identifying the field discriminant with an information volume (Proposition 7.2).
- (C6) **A growth threshold** (§8): an information gain–versus–cost criterion with a *provable* nonzero floor from Smyth’s theorem, and a precise accounting of what is proved versus conjectured.
- (C7) **Applications** (§9): streaming anomaly/novelty detection, a capacity–bounded dictionary learner, and the natural–gradient and minimum–description–length readings of the above.

Honesty discipline. We use *Theorem/Proposition/Lemma* only for statements we prove or verify by exact computation, and *Conjecture/Remark* for model–dependent or heuristic claims (the choice of statistical model, the information exchange rate, the effective Mahler floor, the complex–embedding normalisation). Every displayed numerical value is reproduced by a machine–checked probe (Appendix A).

Related work and positioning. The ingredients are classical. The trace form, Minkowski embedding, different, and discriminant are standard [9, 10]; invariant factors and the rational canonical form are textbook linear algebra [11]; Mahler measure, Northcott’s theorem, and the Lehmer and Smyth bounds are staples of height theory [5, 1, 2, 3, 8]; and the Fisher–information and natural–gradient viewpoint is Amari’s [12, 13]. The contribution is not a new theorem in any of these areas but their *assembly* into an exact, additive learning geometry: a model whose state is a basis of a number field, whose loss is a trace–form residual decided over $\mathbb{Q}$, and whose capacity is bounded by arithmetic height. The trace form plays here the role a kernel plays in kernel methods—a fixed positive form in which all inner products are taken—but it is exact, integer–valued, and carries arithmetic (the discriminant, the different, the height floor) that an analytic kernel does not.

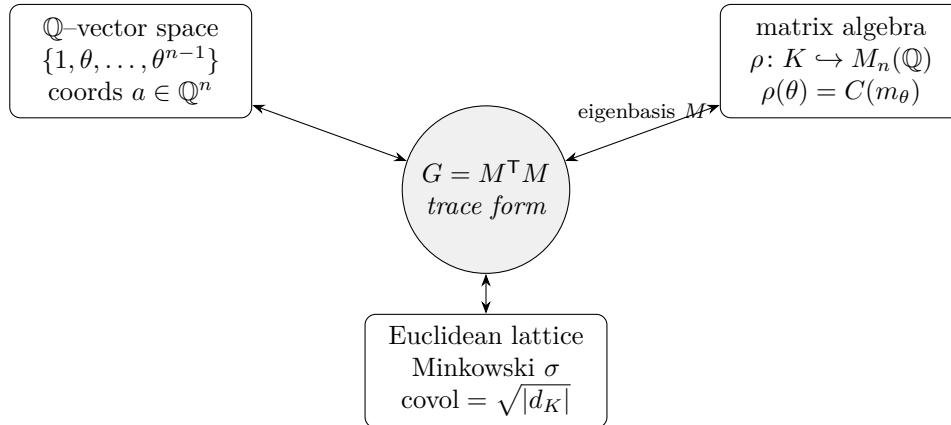


Figure 1: Three views of $K = \mathbb{Q}(\theta)$, glued by the single trace–form Gram G : a $\mathbb{Q}$ –vector space of coordinates, a commutative matrix algebra (the regular representation ρ , diagonalised by the embedding matrix M , Proposition 2.3), and a Euclidean lattice (the Minkowski embedding σ , covolume $\sqrt{|d_K|}$). The same exact rational matrix G serves all three.

Symbol	Meaning
$K = \mathbb{Q}(\theta)$, n	number field of degree $n = [K : \mathbb{Q}]$ with generator θ
(r_1, r_2)	numbers of real / complex-conjugate embeddings, $n = r_1 + 2r_2$
$\mathcal{O}_K$, d_K	ring of integers; field discriminant
m_θ , $C(m_\theta)$	minimal polynomial of θ ; its companion matrix
ρ	regular representation $K \hookrightarrow M_n(\mathbb{Q})$, $\rho(\theta) = C(m_\theta)$
σ_k , M	the n embeddings $K \rightarrow \mathbb{C}$; embedding matrix $M_{ki} = \sigma_k(\omega_i)$
G , T	trace-form Gram $G_{ij} = \text{Tr}(\omega_i \omega_j) = M^\top M$; $T(x, y) = \text{Tr}(xy)$
B , $\text{col } B$	forced basis (the model state); its column span
P , $\mathbf{r}$	trace-form projector onto $\text{col } B$; residual $\mathbf{r} = \mathbf{x} - P\mathbf{x}$
$M(\theta)$, $H(\theta)$	Mahler measure $M = H^{\deg}$; absolute height
μ_S , μ_L	Smyth floor 1.3247... (theorem); Lehmer value 1.17628... (conjectural)
$\mathcal{I}$, λ	Fisher information; information exchange rate (cost of one parameter)

Table 1: Notation used throughout.

2 Preliminaries

2.1 Number fields and the regular representation

Let $K = \mathbb{Q}(\theta)$ with $m_\theta(x) = x^n + c_{n-1}x^{n-1} + \dots + c_0 \in \mathbb{Z}[x]$ irreducible. Write $\mathcal{O}_K$ for the ring of integers. For $x \in K$, multiplication by x is a $\mathbb{Q}$ -linear endomorphism of K ; in the power basis $\{1, \theta, \dots, \theta^{n-1}\}$ it is a matrix $\rho(x) \in M_n(\mathbb{Q})$.

Proposition 2.1 (Regular representation). *The map $\rho: K \rightarrow M_n(\mathbb{Q})$ is an injective $\mathbb{Q}$ -algebra homomorphism, and $\rho(\theta) = C(m_\theta)$, the companion matrix of m_θ . For $x \in K$ with minimal polynomial m_x of degree $d \mid n$, the characteristic and minimal polynomials of the matrix $\rho(x)$ are*

$$\text{charpoly}(\rho(x)) = m_x^{n/d}, \quad \text{minpoly}(\rho(x)) = m_x. \quad (2)$$

Proof. ρ is a ring homomorphism because multiplication in K is associative and bilinear; it is injective because $\rho(x) = 0$ forces $x \cdot 1 = 0$, i.e. $x = 0$. The matrix of multiplication by θ sends $\theta^k \mapsto \theta^{k+1}$ for $k < n-1$ and $\theta^{n-1} \mapsto -\sum_{j < n} c_j \theta^j$, which is exactly $C(m_\theta)$. For (2): K is a vector space over the subfield $\mathbb{Q}(x)$ of degree n/d , and as a $\mathbb{Q}(x)$ -module $K \cong \mathbb{Q}(x)^{n/d}$. Multiplication by x acts on each $\mathbb{Q}(x)$ -summand as multiplication by x , whose matrix over $\mathbb{Q}$ is $C(m_x)$. Hence $\rho(x)$ is similar over $\mathbb{Q}$ to a direct sum of n/d copies of $C(m_x)$, giving $\text{charpoly}(\rho(x)) = m_x^{n/d}$ and $\text{minpoly}(\rho(x)) = m_x$ (the minimal polynomial of a block-diagonal matrix is the least common multiple of the blocks' minimal polynomials, here all equal to m_x). $\square$

Example 2.2 (The regular representation of φ). For φ with $m_\varphi = x^2 - x - 1$ we have $\rho(\varphi) = C(m_\varphi) = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$, and the field trace and norm appear as matrix invariants: $\text{Tr } \rho(\varphi) = 1 = \text{Tr}_{K/\mathbb{Q}}(\varphi)$ and $\det \rho(\varphi) = -1 = N_{K/\mathbb{Q}}(\varphi)$. For $\alpha = \sqrt{5} = 2\varphi - 1$, $\rho(\alpha) = 2\rho(\varphi) - I = \begin{pmatrix} -1 & 2 \\ 2 & 1 \end{pmatrix}$ has $\text{charpoly} = \text{minpoly} = x^2 - 5$. For the rational $\alpha = 3$, $\rho(3) = 3I$ is derogatory: $\text{charpoly} = (x - 3)^2$ but $\text{minpoly} = x - 3$ —the multiplicity $n/d = 2$ of (2) made visible.

2.2 Spectral picture: the regular representation is the conjugates

Proposition 2.3 (Spectral unification). *For every $x \in K$ the eigenvalues of $\rho(x)$ are exactly the n conjugates $\sigma_1(x), \dots, \sigma_n(x)$, and the embedding matrix M diagonalises the regular representation:*

$$M \rho(x) M^{-1} = \text{diag}(\sigma_1(x), \dots, \sigma_n(x)) \quad \text{for all } x \in K. \quad (3)$$

The matrix–algebra and lattice views are thus one operator in two bases: $\rho(x)$ in the power basis, $\text{diag}(\sigma_k(x))$ in the eigenbasis supplied by M .

Proof. For $x \in K$ and a basis element ω_j , $x\omega_j = \sum_i \rho(x)_{ij}\omega_i$; applying the embedding σ_k (a ring homomorphism) gives $\sigma_k(x)\sigma_k(\omega_j) = \sum_i \rho(x)_{ij}\sigma_k(\omega_i)$, that is $\sigma_k(x)M_{kj} = \sum_i M_{ki}\rho(x)_{ij} = (M\rho(x))_{kj}$. Hence $M\rho(x) = \text{diag}(\sigma_k(x))M$, and M is invertible since $\det G = (\det M)^2 = d_K \neq 0$ (Theorem 2.13). Rearranging gives (3), so the eigenvalues of $\rho(x)$ are the diagonal entries $\sigma_k(x)$; equivalently $\text{charpoly}(\rho(x)) = \prod_k (T - \sigma_k(x))$, the field polynomial of x . $\square$

Remark 2.4 (Mahler measure is a spectral invariant). For an algebraic integer θ generating K , $\rho(\theta) = C(m_\theta)$ and (3) identify the spectrum of $\rho(\theta)$ with the conjugates $\{\sigma_k(\theta)\}$, the roots of m_θ . Hence the Mahler measure is read off the spectrum,

$$M(\theta) = \prod_{\lambda \in \text{spec } \rho(\theta), |\lambda| > 1} |\lambda|, \quad (4)$$

the product of the companion eigenvalues outside the unit circle. Equivalently, by Jensen’s formula [5], $M(\theta) = \exp \int_0^1 \log |m_\theta(e^{2\pi it})| dt$. For $\theta = \sqrt{2} + \sqrt{3}$ the spectrum is $\{\pm\sqrt{2} \pm \sqrt{3}\}$ and $M(\theta) = (\sqrt{2} + \sqrt{3})^2 = 5 + 2\sqrt{6}$; for the plastic number (Smyth’s minimiser, the real root of $x^3 - x - 1$) only the real eigenvalue exceeds 1, so $M = \mu_S = 1.3247\dots$ (Figure 3).

Proposition 2.5 (Height as dynamical entropy [17, 7]). *The companion matrix $C(m_\theta) \in M_n(\mathbb{Z})$ defines an endomorphism of the torus $\mathbb{R}^n/\mathbb{Z}^n$, whose topological entropy equals $\log M(\theta)$. For a single endomorphism this is the classical Yuzvinskii–Bowen identity [17]; the Lind–Schmidt–Ward theorem [7] is its $\mathbb{Z}^d$ generalisation, specialising to the present $d = 1$ case. The description length $\log M$ that prices growth in §8 is therefore the entropy of the toral map generated by $\rho(\theta)$ —tying the matrix view (§2.2), the height view (§5), and the threshold (§8) to one invariant.*

Example 2.6 (Diagonalising ρ in $\mathbb{Q}(\sqrt{5})$). For $K = \mathbb{Q}(\sqrt{5})$, $\rho(\varphi) = C(x^2 - x - 1) = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ and $M = \begin{pmatrix} 1 & \varphi \\ 1 & \varphi' \end{pmatrix}$ (rows the two embeddings, $\varphi' = \frac{1-\sqrt{5}}{2}$). Then

$$M \rho(\varphi) M^{-1} = \text{diag}(\varphi, \varphi') = \text{diag}\left(\frac{1+\sqrt{5}}{2}, \frac{1-\sqrt{5}}{2}\right),$$

so the eigenvalues of the companion matrix are exactly the two conjugates and $\text{charpoly}(\rho(\varphi)) = x^2 - x - 1 = m_\varphi$. The same M diagonalises $\rho(x)$ for every $x \in \mathbb{Q}(\sqrt{5})$ at once (Proposition 2.3), and the trace form is the Gram of these eigen–coordinates, $G = M^T M$.

Example 2.7 (The spectrum of $\rho(\sqrt{2} + \sqrt{3})$). For $\theta = \sqrt{2} + \sqrt{3}$, the matrix $\rho(\theta) = C(x^4 - 10x^2 + 1)$ is 4×4 with $\text{charpoly}(\rho(\theta)) = x^4 - 10x^2 + 1 = m_\theta$ and eigenvalues the four conjugates $\{\pm\sqrt{2} \pm \sqrt{3}\} \approx \{\pm 3.146, \pm 0.318\}$. Two lie outside the unit circle, so by (4) $M(\theta) = (\sqrt{2} + \sqrt{3})^2 = 5 + 2\sqrt{6} \approx 9.899$; equivalently $\log M$ is the topological entropy of the toral endomorphism of $C(x^4 - 10x^2 + 1)$ (Proposition 2.5). All values are probe-checked.

2.3 Invariant factors and Smith normal form

The rational canonical form refines (2). For $A \in M_n(\mathbb{Q})$ the *invariant factors* are the monic polynomials $f_1 \mid f_2 \mid \dots \mid f_k$ obtained from the Smith normal form of the characteristic matrix $xI - A$ over the principal ideal domain $\mathbb{Q}[x]$; they satisfy $\prod_i f_i = \text{charpoly}(A)$ and $f_k = \text{minpoly}(A)$, and they form a *complete* invariant for similarity over $\mathbb{Q}$. We will use:

Corollary 2.8. *For $\alpha \in K = \mathbb{Q}(\theta)$, the minimal polynomial m_α equals the largest invariant factor of $xI - \rho(\alpha)$.*

Proof. Immediate from $f_k = \text{minpoly}(\rho(\alpha)) = m_\alpha$ by Proposition 2.1. $\square$

Theorem 2.9 (Completeness of invariant factors). *Matrices $A, A' \in M_n(\mathbb{Q})$ are similar over $\mathbb{Q}$ if and only if they have the same invariant factors. The characteristic polynomial alone is a complete similarity invariant only on the non-derogatory locus (where $\text{charpoly} = \text{minpoly}$); in general it is not.*

Proof. $A \sim A'$ over $\mathbb{Q}$ iff the $\mathbb{Q}[x]$ -modules $\mathbb{Q}[x]^n/(xI - A)$ and $\mathbb{Q}[x]^n/(xI - A')$ are isomorphic, iff $xI - A$ and $xI - A'$ have the same Smith normal form, i.e. the same invariant factors; this is the structure theorem for finitely generated modules over the PID $\mathbb{Q}[x]$ [11, Ch. 12]. The non-derogatory case is $k = 1$, where the single invariant factor equals the characteristic polynomial. $\square$

Example 2.10 (Characteristic polynomial is insufficient). In $K = \mathbb{Q}(\varphi)$ compare the block sum $\rho(\varphi) \oplus \rho(\varphi) = C(m_\varphi) \oplus C(m_\varphi)$ with the companion $C((x^2 - x - 1)^2)$ of $(x^2 - x - 1)^2 = x^4 - 2x^3 - x^2 + 2x + 1$. Both are 4×4 with characteristic polynomial $(x^2 - x - 1)^2$ and trace 2, yet they are *not* similar: the former has minimal polynomial $x^2 - x - 1$ and invariant factors $(x^2 - x - 1, x^2 - x - 1)$, the latter the single invariant factor $(x^2 - x - 1)^2$. The largest invariant factor separates them; the characteristic polynomial does not. This is precisely why Theorem 4.2 reads the minimal polynomial as the largest invariant factor.

Remark 2.11 (Computing the Smith normal form). Over the Euclidean domain $\mathbb{Q}[x]$ the invariant factors are obtained from $xI - A$ by the standard reduction: repeatedly select a nonzero entry of least degree as pivot, clear its row and column by polynomial division (re-pivoting whenever a smaller-degree remainder surfaces), and then enforce divisibility of the trailing submatrix; the resulting monic diagonal entries of degree ≥ 1 are $f_1 \mid \cdots \mid f_k$. The procedure is exact over $\mathbb{Q}$ and terminates because the minimal pivot degree strictly decreases.

Example 2.12 (Why the full list is needed). Even the pair $(\text{charpoly}, \text{minpoly})$ is incomplete once an invariant factor has multiplicity ≥ 3 . With $J_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $J_1 = (0)$, the matrices $J_2 \oplus J_2$ and $J_2 \oplus J_1 \oplus J_1$ both have characteristic polynomial x^4 and minimal polynomial x^2 , yet they are *not* similar: their invariant factors are (x^2, x^2) and (x, x, x^2) . Only the full invariant-factor list separates them—one reason the bridge (Theorem 4.2) reads the largest invariant factor from the Smith normal form rather than from the characteristic polynomial.

This is the exact algorithm underlying our coordinates-to-minimal-polynomial bridge (§4.3); it is purely rational and never computes a root.

2.4 The Minkowski embedding and the trace form

Let $\sigma_1, \dots, \sigma_{r_1} : K \rightarrow \mathbb{R}$ be the real embeddings and $\sigma_{r_1+1}, \bar{\sigma}_{r_1+1}, \dots$ the complex pairs, $n = r_1 + 2r_2$. The Minkowski embedding is $\sigma(x) = (\sigma_1(x), \dots, \sigma_n(x))$. The trace is the sum of conjugates, $\text{Tr}_{K/\mathbb{Q}}(x) = \sum_i \sigma_i(x)$, whence (1). In a basis $\{\omega_i\}$ define the Gram matrix and the embedding matrix

$$G_{ij} = \text{Tr}(\omega_i \omega_j), \quad M_{ki} = \sigma_k(\omega_i) \in \mathbb{C}. \quad (5)$$

Theorem 2.13 (Trace form as embedding Gram). *$G = M^\top M$. In particular, if K is totally real ($r_2 = 0$) then M is real and G is a genuine Gram matrix of n vectors in $\mathbb{R}^n$. For an integral basis of $\mathcal{O}_K$, $\det G = \text{disc}(K) = d_K$; for a sub-order $\Lambda \subseteq \mathcal{O}_K$, $\det G = [\mathcal{O}_K : \Lambda]^2 d_K$. The covolume of the Minkowski lattice is treated in §2.6.*

Proof. $G_{ij} = \text{Tr}(\omega_i \omega_j) = \sum_k \sigma_k(\omega_i \omega_j) = \sum_k \sigma_k(\omega_i) \sigma_k(\omega_j) = \sum_k M_{ki} M_{kj} = (M^\top M)_{ij}$, using that each σ_k is a ring homomorphism. The discriminant identity $\det G = d_K$ for an integral basis and the index correction are standard [9, 10]; the relation $\det G = (\det M)^2$ yields the lattice covolume of §2.6. $\square$

Remark 2.14 (Signature and totally real fields). For $r_2 > 0$ the real quadratic form $\sum_i \sigma_i(x)^2$ has signature $(r_1 + r_2, r_2)$; the positive-definite companion is the Hermitian trace form of §2.5. Apart from that subsection's Gaussian-integer example, all worked examples here are totally real, where $G = M^\top M$ is itself positive definite and no normalisation subtlety arises.

Example 2.15 (The golden field). Let $K = \mathbb{Q}(\sqrt{5}) = \mathbb{Q}(\varphi)$, $\varphi = \frac{1+\sqrt{5}}{2}$, $m_\varphi = x^2 - x - 1$, so $\rho(\varphi) = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$. The embeddings send $\varphi \mapsto \varphi$ and $\varphi \mapsto \varphi' = \frac{1-\sqrt{5}}{2}$, so in the basis $\{1, \varphi\}$,

$$M = \begin{pmatrix} 1 & \varphi \\ 1 & \varphi' \end{pmatrix}, \quad G = M^\top M = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \quad \det G = 5 = d_{\mathbb{Q}(\sqrt{5})}.$$

Here $\text{Tr}(1) = 2$, $\text{Tr}(\varphi) = \varphi + \varphi' = 1$, $\text{Tr}(\varphi^2) = \varphi^2 + \varphi'^2 = 3$, and $\{1, \varphi\}$ is an integral basis ($\mathbb{Z}[\varphi] = \mathcal{O}_K$).

Example 2.16 (A biquadratic field). Let $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ with the product basis $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$. The four embeddings send $(\sqrt{2}, \sqrt{3}) \mapsto (\pm\sqrt{2}, \pm\sqrt{3})$ (four sign choices), and

$$G = \text{diag}(4, 8, 12, 24), \quad \det G = 9216 = 2^{10} 3^2.$$

Since $d_{\mathbb{Q}(\sqrt{2}, \sqrt{3})} = 2304 = 2^8 3^2$, this basis spans a sub-order of index $[\mathcal{O}_K : \Lambda] = \sqrt{9216/2304} = 2$; the Gram and all geometry below are taken relative to this convenient basis.

2.5 Complex embeddings and the positive-definite form

When $r_2 > 0$ the trace form $T(x, y) = \sum_i \sigma_i(x) \sigma_i(y)$ is *indefinite*: a complex place contributes, through its conjugate pair $\sigma, \bar{\sigma}$, the term $\sigma(x) \sigma(y) + \bar{\sigma}(x) \bar{\sigma}(y) = 2\Re(\sigma(x) \sigma(y))$, a hyperbolic plane, and the signature of T is $(r_1 + r_2, r_2)$. The positive-definite companion is the *Hermitian trace form*

$$T_2(x, y) = \sum_{i=1}^n \sigma_i(x) \overline{\sigma_i(y)} = \sum_{i \leq r_1} \sigma_i(x) \sigma_i(y) + 2 \sum_{j=1}^{r_2} \Re(\sigma_{r_1+j}(x) \overline{\sigma_{r_1+j}(y)}), \quad (6)$$

the pullback of the canonical (Minkowski) inner product on $\mathbb{R}^{r_1} \times \mathbb{C}^{r_2} \cong \mathbb{R}^n$. Its Gram matrix is $G_2 = M^* M$ (M^* the conjugate transpose), positive definite, with $\det G_2 = |d_K|$ for an integral basis, so the Minkowski lattice has covolume $\sqrt{\det G_2} = \sqrt{|d_K|}$ in this metric (the geometric embedding $\mathbb{C} \cong \mathbb{R}^2$ rescales the complex coordinates to the textbook value $2^{-r_2} \sqrt{|d_K|}$). For totally real fields $r_2 = 0$ and $T_2 = T$, $G_2 = G = M^\top M$ (Theorem 2.13); all of §3–7 is written for G and applies verbatim to G_2 in the complex case, with $M^\top$ replaced by M^* and the isotropic Gaussian of §7 taken on the canonical real space $\mathbb{R}^n$ with metric G_2 .

Example 2.17 (The Gaussian field). $K = \mathbb{Q}(i)$ ($r_1 = 0$, $r_2 = 1$, $n = 2$), basis $\{1, i\}$, embeddings $i \mapsto i$ and $i \mapsto -i$. The trace form is

$$G = \begin{pmatrix} \text{Tr}(1) & \text{Tr}(i) \\ \text{Tr}(i) & \text{Tr}(i^2) \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix},$$

indefinite of signature $(1, 1)$, while the Hermitian form (6) is $G_2 = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = 2I$, positive definite, with $\det G_2 = 4 = |d_{\mathbb{Q}(i)}|$ so the covolume in this metric is $\sqrt{4} = 2 = \sqrt{|d_K|}$; the geometric rescaling returns the unit-covolume Gaussian lattice $\mathbb{Z}[i]$ ($2^{-1} \sqrt{4} = 1$). Membership and residual are taken with respect to G_2 .

Example 2.18 (A cubic with a complex place). $K = \mathbb{Q}(\sqrt[3]{2})$ has $r_1 = 1, r_2 = 1, n = 3$; the embeddings send $\sqrt[3]{2}$ to the real $\sqrt[3]{2}$ and to the complex pair $\sqrt[3]{2}\omega, \sqrt[3]{2}\omega^2$ ($\omega = e^{2\pi i/3}$). On the power basis $\{1, \sqrt[3]{2}, \sqrt[3]{4}\}$, using $\text{Tr}(1) = 3$ and $\text{Tr}(\sqrt[3]{2}) = \text{Tr}(\sqrt[3]{4}) = 0$,

$$G = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 0 & 6 \\ 0 & 6 & 0 \end{pmatrix}, \quad \det G = -108 = d_{\mathbb{Q}(\sqrt[3]{2})}.$$

This G is *indefinite*, of signature $(2, 1) = (r_1 + r_2, r_2)$ (eigenvalues $3, \pm 6$), exactly as above; the positive-definite companion is the Hermitian $G_2 = M^*M$. The exact identity $\det G = d_K$ persists through the complex place.

2.6 The Minkowski lattice

By Theorem 2.13 the embedding σ sends $\mathcal{O}_K$ to a full-rank lattice $\Lambda_K = \sigma(\mathcal{O}_K) \subset \mathbb{R}^n$ with Gram matrix G . Its covolume is $\text{covol}(\Lambda_K) = \sqrt{\det G} = \sqrt{|d_K|}$ ($= \sqrt{d_K}$ for a totally real field; the geometric normalisation of the complex places gives $2^{-r_2}\sqrt{|d_K|}$ instead, §2.5). For $\mathbb{Q}(\sqrt{5})$, $\det G = 5$ and the covolume is $\sqrt{5}$. Lattice and algebra thus share a single matrix G : distances, volumes, and the trace pairing are all computed by the same exact rational form, and the “forced” directions of the learner (§4) are a sublattice spanned by a chosen basis.

Remark 2.19 (The lattice meets the height floor). The Mahler measure that gates capacity (§5) is a function of where the conjugates of θ sit, hence of the geometry of Λ_K . Lehmer’s problem—whether $M(\theta) \geq \mu_L > 1$ for some universal μ_L and every non-cyclotomic θ —is, in this picture, a question of a universal lower bound on a height attached to lattice points, and Smyth’s unconditional floor $\mu_S = 1.3247\dots$ (Theorem 8.2) is exactly the provable fragment we use. The growth threshold of §8 thus inherits a genuinely arithmetic—and, beyond Smyth’s case, conjectural—lower bound from the lattice.

K	(r_1, r_2)	$\det G = d_K$	$\text{covol} = \sqrt{ d_K }$	different $\mathfrak{d}$	$N(\mathfrak{d})$
$\mathbb{Q}(\sqrt{5})$	(2, 0)	5	$\sqrt{5}$	$(\sqrt{5})$	5
$\mathbb{Q}(\sqrt{2})$	(2, 0)	8	$2\sqrt{2}$	$(2\sqrt{2})$	8
$\mathbb{Q}(\sqrt{3})$	(2, 0)	12	$2\sqrt{3}$	$(2\sqrt{3})$	12
$\mathbb{Q}(\sqrt{7})$	(2, 0)	28	$2\sqrt{7}$	$(2\sqrt{7})$	28
$\mathbb{Q}(i)$	(0, 1)	−4	2	$(2i)$	4
$\mathbb{Q}(\sqrt[3]{2})$	(1, 1)	−108	$6\sqrt{3}$	$(3\sqrt[3]{4})$	108

Table 2: A catalog of fields, every entry reproduced by the probe (§A). For an integral basis $\det G = d_K$; the covolume is $\sqrt{|d_K|}$ (the Fisher–Jeffreys cell volume, Proposition 7.2); and the different $\mathfrak{d} = (m'_\theta(\theta))$ satisfies $N(\mathfrak{d}) = |d_K|$. For $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt[3]{2})$ the trace form is indefinite of signature $(r_1 + r_2, r_2)$, and the covolume uses the Hermitian companion G_2 (§2.5).

We close the preliminaries by recording the three views of K that the construction uses interchangeably.

View	Structure	Realisation in $K = \mathbb{Q}(\theta)$
Vector space	$\mathbb{Q}$ -basis $\{1, \theta, \dots, \theta^{n-1}\}$	coordinates $a \in \mathbb{Q}^n$
Matrix algebra	regular representation $\rho: K \hookrightarrow M_n(\mathbb{Q})$	$\rho(\theta) = C(m_\theta)$
Euclidean lattice	Minkowski embedding σ , Gram $G = M^\top M$	$\text{covol} = \sqrt{ d_K }$

The three views share one matrix G : the same exact rational form measures coordinates, matrix products, and lattice distances. The remainder of the paper is the learning theory of this single object.

3 The trace–form geometry

3.1 The projector and the residual

Fix a “forced” basis: a full–rank matrix $B \in \mathbb{Q}^{n \times m}$ whose columns are coordinate vectors of algebraic integers in K . Write $\text{col}(B)$ for the G –subspace they span.

Definition 3.1 (Trace–form projector and residual). Provided $B^\top G B$ is invertible, set

$$P = B(B^\top G B)^{-1} B^\top G, \quad \mathbf{r}(\mathbf{x}) = \mathbf{x} - P\mathbf{x}. \quad (7)$$

If $B^\top G B$ is singular we write $P = \text{NO_PROJECTION}$.

Proposition 3.2 (Idempotence, orthogonality, capture). *With P as in (7): (i) $P^2 = P$; (ii) $B^\top G \mathbf{r}(\mathbf{x}) = 0$, i.e. $\mathbf{r}(\mathbf{x})$ is G –orthogonal to $\text{col}(B)$; (iii) $P\mathbf{x} = \mathbf{x}$ iff $\mathbf{x} \in \text{col}(B)$; and (iv) the exact membership criterion holds:*

$$\mathbf{x} \in \text{col}(B) \iff \mathbf{r}(\mathbf{x}) = 0 \iff \|\mathbf{r}(\mathbf{x})\|_G^2 = \mathbf{r}^\top G \mathbf{r} = 0. \quad (8)$$

Proof. $P^2 = B(B^\top G B)^{-1}(B^\top G B)(B^\top G B)^{-1} B^\top G = B(B^\top G B)^{-1} B^\top G = P$. For (ii), $B^\top G \mathbf{r} = B^\top G \mathbf{x} - (B^\top G B)(B^\top G B)^{-1} B^\top G \mathbf{x} = 0$. For (iii), if $\mathbf{x} = B\mathbf{c}$ then $P\mathbf{x} = B(B^\top G B)^{-1}(B^\top G B)\mathbf{c} = B\mathbf{c} = \mathbf{x}$; conversely $P\mathbf{x} = \mathbf{x}$ lies in $\text{col}(B)$. (iv) follows since G is positive definite on a totally real field, so $\mathbf{r}^\top G \mathbf{r} = 0 \iff \mathbf{r} = 0$. $\square$

Example 3.3 (A projector in the golden field). In $K = \mathbb{Q}(\sqrt{5})$ with $G = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$, take the forced basis $B = (0, 1)^\top$ (the line $\mathbb{Q}\varphi$). Then $B^\top G B = 3$ and

$$P = B(B^\top G B)^{-1} B^\top G = \frac{1}{3} \begin{pmatrix} 0 & 0 \\ 1 & 3 \end{pmatrix}.$$

The element $2\varphi = (0, 2)^\top$ satisfies $P(0, 2)^\top = (0, 2)^\top$, residual 0: captured. The element $\sqrt{5} = 2\varphi - 1 = (-1, 2)^\top$ gives $P\sqrt{5} = (0, \frac{5}{3})^\top$ and $\mathbf{r} = (-1, \frac{1}{3})^\top \neq 0$, with $\|\mathbf{r}\|_G^2 = \frac{5}{3} > 0$: off–axis novelty. Every quantity is an exact rational.

Remark 3.4 (Exact membership, real magnitude). Criterion (8) is decided in $\mathbb{Q}$ (all of $B, G, P, \mathbf{r}$ are rational). The number $\|\mathbf{r}\|_G$ is used only as a real *magnitude* (“how far off”). We never threshold a floating residual to decide membership; this discipline—decide over $\mathbb{Q}/\mathbb{Z}$, measure over $\mathbb{R}$ —runs through the entire construction.

Example 3.5 (A complex–embedding projector). We carry $\mathbb{Q}(i)$ through the projector with its Hermitian Gram $G_2 = 2I$ (§2.5, positive definite). With forced basis $B = \langle 1 \rangle = (1, 0)^\top$ and target $x = i = (0, 1)^\top$, one has $B^* G_2 B = 2$ and $B^* G_2 x = 0$, so $Px = 0$ and $\mathbf{r} = i$ with $\|\mathbf{r}\|_{G_2}^2 = (0, 1) G_2 (0, 1)^\top = 2$. The residual is all of i : it is G_2 –orthogonal to 1 (indeed $\text{Tr}_{\mathbb{Q}(i)/\mathbb{Q}}(i) = 0$), so i is pure off–axis novelty against $\langle 1 \rangle$. The exact projector/residual calculus thus runs verbatim over the Hermitian form when $r_2 > 0$; what remains open (O2) is the probe–level verification of the Fisher identities in the genuinely complex case, which we flag rather than assert.

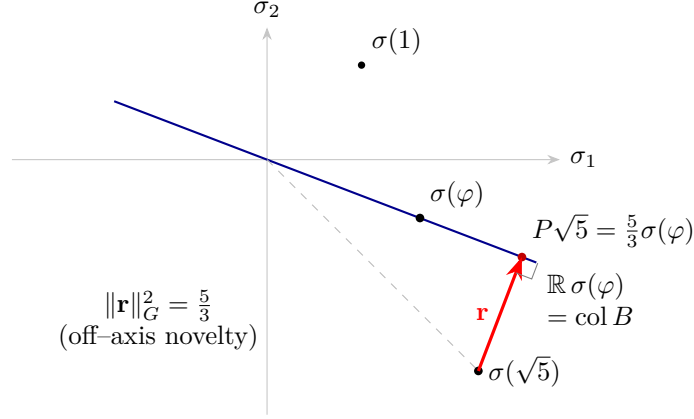


Figure 2: The projector and residual of Examples 3.3 and 7.12, drawn in the Minkowski plane of $\mathbb{Q}(\sqrt{5})$ where the trace metric G is the Euclidean metric. The forced line is $\text{col } B = \mathbb{R} \sigma(\varphi)$; the point $\sigma(\sqrt{5})$ projects G -orthogonally to $P\sqrt{5} = \frac{5}{3} \sigma(\varphi)$, and the perpendicular drop is the residual $\mathbf{r}$ with $\|\mathbf{r}\|_G^2 = \frac{5}{3}$.

3.2 Understanding as zero residual

Equation (8) is the framework's operational definition of *captured* (“understood”): $\mathbf{x}$ is explained by the current dictionary exactly when $\mathbf{r}(\mathbf{x}) = 0$. A persistently nonzero residual is the signal that the dictionary is too small—the object to be learned. The orthogonality $B^\top G \mathbf{r} = 0$ guarantees that the residual carries no component already in the forced subspace; it is pure novelty.

3.3 Trace duality: dual basis, codifferent, and different

The inverse Gram G^{-1} in the projector $P = B(B^\top G B)^{-1} B^\top G$ is not an analytic convenience; it is the matrix of *trace duality*. For a basis $\{\omega_i\}$ of K the dual basis $\{\omega_i^\vee\}$ is defined by $\text{Tr}(\omega_i \omega_j^\vee) = \delta_{ij}$, and in coordinates $\omega_i^\vee = \sum_j (G^{-1})_{ij} \omega_j$, since $\text{Tr}(\omega_i \omega_j^\vee) = \sum_k (G^{-1})_{jk} G_{ki} = (G^{-1} G)_{ji} = \delta_{ij}$. The lattice $L = \bigoplus_i \mathbb{Z} \omega_i$ has *dual lattice*

$$L^\vee = \{x \in K : \text{Tr}(xL) \subseteq \mathbb{Z}\} = \bigoplus_i \mathbb{Z} \omega_i^\vee,$$

and for $L = \mathcal{O}_K$ this is the *codifferent* $\mathcal{O}_K^\vee \supseteq \mathcal{O}_K$, a fractional ideal whose inverse is the *different* $\mathfrak{d} = (\mathcal{O}_K^\vee)^{-1} \subseteq \mathcal{O}_K$. Trace duality binds the metric to the discriminant: for an integral basis $\det G = d_K$, and the different computes the discriminant as a norm, $N(\mathfrak{d}) = |d_K|$. For a non-maximal order $\mathbb{Z}[\theta] \subseteq \mathcal{O}_K$ the trace-form determinant carries the index,

$$\text{disc}(\mathbb{Z}[\theta]) = [\mathcal{O}_K : \mathbb{Z}[\theta]]^2 d_K,$$

so $\det G$ in a power basis already measures how far θ is from generating the full ring of integers.

Example 3.6 (Trace duality in $\mathbb{Q}(\sqrt{5})$). With basis $\{1, \varphi\}$, $G = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$, $\det G = 5 = d_{\mathbb{Q}(\sqrt{5})}$ and $G^{-1} = \frac{1}{5} \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}$. The dual basis is $1^\vee = \frac{1}{5}(3 - \varphi)$ and $\varphi^\vee = \frac{1}{5}(2\varphi - 1)$; one checks $\text{Tr}(1 \cdot 1^\vee) = \frac{1}{5}(3 \text{Tr } 1 - \text{Tr } \varphi) = \frac{1}{5}(6 - 1) = 1$ and $\text{Tr}(\varphi \cdot 1^\vee) = \frac{1}{5}(3 \text{Tr } \varphi - \text{Tr } \varphi^2) = \frac{1}{5}(3 - 3) = 0$. The different is $\mathfrak{d} = (m'(\varphi)) = (2\varphi - 1) = (\sqrt{5})$ with $N(\mathfrak{d}) = 5 = |d_K|$, and $\mathcal{O}_K^\vee = \frac{1}{\sqrt{5}} \mathcal{O}_K$.

Remark 3.7 (Why this matters for learning). The residual $\mathbf{r} = x - Px$ is the component of x that pairs to zero, under Tr , against the whole forced subspace—the part of x *invisible to the current dictionary in the trace pairing*. Capturing it (§4) adjoins the missing dual direction. Projector, residual, and dual lattice are thus three views of one object: the trace form and its inverse.

Remark 3.8 (Galois invariance of capture). The trace pairing is invariant under every automorphism g of $K/\mathbb{Q}$: since g permutes the embeddings σ_k , we have $\text{Tr}(g(x)g(y)) = \text{Tr}(g(xy)) = \text{Tr}(xy)$. Hence if $K/\mathbb{Q}$ is Galois, $\text{Gal}(K/\mathbb{Q})$ acts by G -isometries ($g^\top Gg = G$ on coordinates); a Galois-stable forced subspace has a Galois-stable residual, and “captured” ($\mathbf{r} = 0$) is a Galois-invariant notion. The learner’s geometry respects the arithmetic symmetry of the field.

4 Learning as exact basis growth

4.1 The residual is the loss

In place of a differentiable loss minimised by gradient descent, the framework uses the exact residual as its error signal. The Pythagorean decomposition

$$\|\mathbf{x}\|_G^2 = \|P\mathbf{x}\|_G^2 + \|\mathbf{r}(\mathbf{x})\|_G^2 \quad (9)$$

(immediate from G -orthogonality of $P\mathbf{x}$ and $\mathbf{r}$) splits each observation into a captured part and a residual; learning drives the residual to zero by enlarging $\text{col}(B)$.

4.2 The propose-for-confirm protocol and guardrails

Growth is disciplined. We summarise the invariants that the implementation maintains; they are not theorems about K but design guarantees that keep the learner exact and auditable.

Protocol 4.1 (Gated, witnessed growth). A streaming learner maintains a forced basis B , an exact residual-field accumulator (a running mean of $\mathbf{r}$ over consecutive off-axis observations), and a persistence counter. It exposes three operations:

- (a) *observe*($\mathbf{x}$): compute $\mathbf{r}(\mathbf{x})$ and update the accumulator; never mutates B ;
- (b) *propose*(): when the residual centroid has persisted and a capacity gate (§5) admits the implied seed, return a candidate algebraic-integer seed; pure;
- (c) *confirm*(seed): the *sole* mutator—adjoin the seed’s coordinate vector to B , re-derive P , and append the growth to a SHA-256 witness chain.

The maintained invariants are: (*exactness*) every quantity entering a membership or admissibility decision is a $\mathbb{Q}/\mathbb{Z}$ value, floats appearing only in display; (*algebraic integrality*) every seed is a monic-integer minimal polynomial; (*sole mutator*) only *confirm* changes B ; (*witnessed*) each growth is a tamper-evident, replayable record. These are the guardrails that let one trust the learner’s output bit-for-bit.

4.3 The coordinates-to-minimal-polynomial bridge

A residual centroid is a coordinate vector; to adjoin it as a named algebraic-integer seed we need its minimal polynomial. We compute it exactly.

Theorem 4.2 (Exact seed identity). *Let $\alpha = \sum_i a_i \theta^i \in K$ with $a_i \in \mathbb{Q}$, and let $M_\alpha = \rho(\alpha) = \sum_i a_i C(m_\theta)^i$ be its matrix under ρ . Then m_α , the minimal polynomial of α over $\mathbb{Q}$, equals the largest invariant factor of $xI - M_\alpha$, computed by Smith normal form over $\mathbb{Q}[x]$. If $\alpha \in \mathcal{O}_K$ (an algebraic integer) then $m_\alpha \in \mathbb{Z}[x]$ is monic; otherwise the computation produces a non-integer monic polynomial and the seed is rejected.*

Proof. M_α is exactly the matrix of multiplication by α , so $\text{minpoly}(M_\alpha) = m_\alpha$ by Proposition 2.1, and this is the largest invariant factor by Corollary 2.8. Integrality of m_α for $\alpha \in \mathcal{O}_K$ is the defining property of algebraic integers; the converse guards the seed dictionary against non-integer elements. $\square$

Example 4.3 (The bridge, worked). In $K = \mathbb{Q}(\theta)$, $\theta = \sqrt{2} + \sqrt{3}$, $m_\theta = x^4 - 10x^2 + 1$, the companion matrix and the regular-representation of $2\sqrt{6} = \theta^2 - 5$ (coordinate vector $(-5, 0, 1, 0)$) are

$$\rho(\theta) = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 10 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad \rho(2\sqrt{6}) = \rho(\theta)^2 - 5I = \begin{pmatrix} -5 & 0 & -1 & 0 \\ 0 & -5 & 0 & -1 \\ 1 & 0 & 5 & 0 \\ 0 & 1 & 0 & 5 \end{pmatrix}.$$

Reducing $xI - \rho(2\sqrt{6})$ to Smith normal form over $\mathbb{Q}[x]$ gives $\text{diag}(1, 1, x^2 - 24, x^2 - 24)$, so the invariant factors are $(x^2 - 24, x^2 - 24)$ and the largest—hence $m_{2\sqrt{6}}$ —is $x^2 - 24$. This matches the elementary check $(2\sqrt{6})^2 = 24$, and the doubled factor reflects $n/d = 4/2 = 2$ (Proposition 2.1). The polynomial is monic over $\mathbb{Z}$, so $2\sqrt{6}$ is admitted as an algebraic-integer seed.

4.4 Cross-field growth and the compositum

Some residuals are not elements of K at all: the observed algebraic quantity is forced into a strictly larger field. Capturing it requires not a new vector but a new *field*. We work in the *disjoint* (linearly disjoint) case.

Definition 4.4 (Linear disjointness). Finite extensions $K, L/\mathbb{Q}$ are linearly disjoint if $K \otimes_{\mathbb{Q}} L \rightarrow KL$ is injective, equivalently $[KL : \mathbb{Q}] = [K : \mathbb{Q}][L : \mathbb{Q}]$, equivalently $\{e_i f_j\}$ is a $\mathbb{Q}$ -basis of KL for $\mathbb{Q}$ -bases $\{e_i\}$ of K and $\{f_j\}$ of L .

Proposition 4.5 (Kronecker Gram of a disjoint compositum). *If K, L are linearly disjoint with bases $\{e_i\}$ ($i \leq m$) and $\{f_j\}$ ($j \leq k$), then the trace-form Gram of KL on the product basis $\{e_i f_j\}$ factors as*

$$G_{KL} = G_K \otimes G_L, \quad \det G_{KL} = (\det G_K)^k (\det G_L)^m. \quad (10)$$

Proof. The embeddings of the disjoint compositum are exactly the products $\tau \otimes v$ of an embedding τ of K and an embedding v of L , and on $a \in K$, $b \in L$ one has $(\tau \otimes v)(ab) = \tau(a)v(b)$. Hence for $a \in K$, $b \in L$,

$$\text{Tr}_{KL/\mathbb{Q}}(ab) = \sum_{\tau, v} \tau(a)v(b) = \left(\sum_{\tau} \tau(a) \right) \left(\sum_v v(b) \right) = \text{Tr}_{K/\mathbb{Q}}(a) \text{Tr}_{L/\mathbb{Q}}(b).$$

Applying this to $a = e_i e_{i'}$, $b = f_j f_{j'}$ gives

$$\begin{aligned} (G_{KL})_{(i,j),(i',j')} &= \text{Tr}_K(e_i e_{i'}) \text{Tr}_L(f_j f_{j'}) \\ &= (G_K)_{ii'} (G_L)_{jj'}, \end{aligned}$$

which is the Kronecker product. The determinant in (10) is the standard identity

$$\det(A \otimes B) = (\det A)^{\dim B} (\det B)^{\dim A}, \quad \dim G_K = m, \dim G_L = k.$$

□

Corollary 4.6 (Multiquadratic tower). *For pairwise linearly disjoint $K_1, \dots, K_r$, iterating (10) gives $G_{K_1 \dots K_r} = G_{K_1} \otimes \dots \otimes G_{K_r}$. In particular $\mathbb{Q}(\sqrt{p_1}, \dots, \sqrt{p_r})$ for distinct primes p_i has, on the 2^r squarefree-product basis, $G = \bigotimes_{i=1}^r \begin{pmatrix} 2 & 0 \\ 0 & 2p_i \end{pmatrix}$ —a diagonal matrix of order 2^r with entries $2^r \prod_{i \in S} p_i$ over subsets $S \subseteq \{1, \dots, r\}$. For $\{p_i\} = \{2, 3\}$ this is $\text{diag}(4, 8, 12, 24)$, recovering Example 2.16.*

Example 4.7 (Adjoining $\sqrt{7}$). $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ ($G_K = \text{diag}(4, 8, 12, 24)$) and $L = \mathbb{Q}(\sqrt{7})$ ($G_L = \text{diag}(2, 14)$) are linearly disjoint, since $[KL : \mathbb{Q}] = 8 = 4 \cdot 2$. By (10),

$$G_{KL} = G_K \otimes G_L = \text{diag}(8, 56, 16, 112, 24, 168, 48, 336), \quad \det G_{KL} = 9216^2 \cdot 28^4.$$

After adjoining $\sqrt{7}$ the previously out-of-field residual is captured and its field residual returns to 0.

Remark 4.8 (The non-disjoint case is open). When the candidate generator's minimal polynomial *partially factors* over K , the compositum is strictly smaller than the tensor product, $[K(\beta) : K] < \deg m_\beta$, and (10) no longer applies; one must build the true compositum basis (e.g. via a primitive element and resultants, with factorisation over a number field). This is a genuine open extension of the present work; the implemented learner detects it and refuses growth rather than producing a fictitious Gram.

Example 4.9 (A non-disjoint extension). Over $K = \mathbb{Q}(\sqrt{2})$ adjoin $\beta = \sqrt{2} + \sqrt{3}$, whose minimal polynomial over $\mathbb{Q}$ is $x^4 - 10x^2 + 1$. Over K this polynomial *factors*:

$$x^4 - 10x^2 + 1 = (x^2 + 2\sqrt{2}x - 1)(x^2 - 2\sqrt{2}x - 1) \quad \text{in } \mathbb{Q}(\sqrt{2})[x],$$

since $2\sqrt{2} \in K$ (one checks $\beta^2 - 2\sqrt{2}\beta - 1 = 0$). Hence $[K(\beta) : K] = 2 < 4 = \deg m_\beta$, so K and $\mathbb{Q}(\beta)$ are *not* linearly disjoint. The true compositum $K(\beta) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ has degree 4, not the tensor degree $[K : \mathbb{Q}] \deg m_\beta = 2 \cdot 4 = 8$; the Kronecker Gram $G_K \otimes G_{\mathbb{Q}(\beta)}$ would describe an 8-dimensional space that does not exist. The learner detects the partial factorisation and refuses growth (Remark 4.8); constructing the correct degree-4 Gram requires a primitive element of $K(\beta)$ and is left open.

5 Capacity control via heights

Unbounded basis growth would defeat the purpose of an exact, finite model. We bound capacity by arithmetic height, which makes the admissible seed set provably finite.

5.1 Mahler measure and height

Definition 5.1 (Mahler measure). For $p(x) = a_n \prod_i (x - \alpha_i) \in \mathbb{C}[x]$, $M(p) = |a_n| \prod_i \max(1, |\alpha_i|)$. For an algebraic θ with minimal polynomial m_θ over $\mathbb{Z}$ (primitive integer form), $M(\theta) = M(m_\theta)$.

Theorem 5.2 (Height, Landau, Northcott). *Let $p \in \mathbb{Z}[x]$ be monic of degree d with coefficient vector $(1, c_{d-1}, \dots, c_0)$.*

(a) $M(\theta) = H(\theta)^{\deg \theta}$, where H is the absolute multiplicative Weil height [8].

(b) (Landau) $M(p) \leq \|p\|_2 = (\sum_i c_i^2)^{1/2}$, hence $M(p)^2 \leq \sum_i c_i^2 \in \mathbb{Z}$.

(c) (Northcott) For each D, B there are only finitely many algebraic integers θ with $\deg \theta \leq D$ and $M(\theta) \leq B$ [1].

Proof. (a) is the definition of the Weil height via local contributions [8]. (b) is Landau's inequality [6, 5]. (c): a monic integer polynomial of degree $\leq D$ with $M \leq B$ has, by Mahler's coefficient bound $|c_j| \leq \binom{d}{j} M(p)$, bounded integer coefficients, so there are finitely many such polynomials, hence finitely many such θ . $\square$

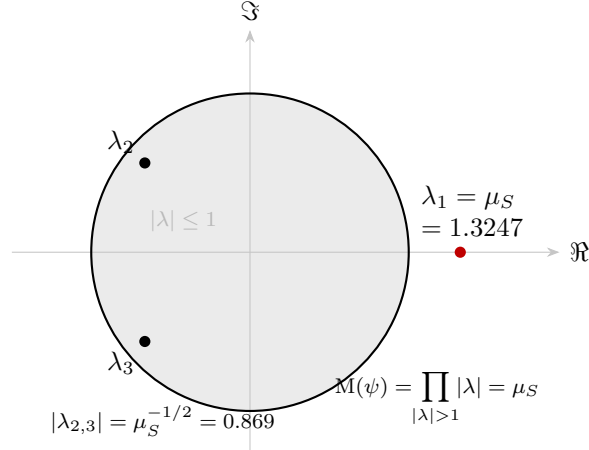


Figure 3: The Mahler picture for the plastic number ψ (Smyth's minimiser, root of $x^3 - x - 1$): the eigenvalues of $\rho(\psi) = C(x^3 - x - 1)$ plotted in $\mathbb{C}$ with the unit circle. Only $\lambda_1 = \psi$ lies outside, so $M(\psi) = \prod_{|\lambda|>1} |\lambda| = \mu_S = 1.3247\dots$ (Remark 2.4). Smyth's bound μ_S is a theorem for non-reciprocal integers (Theorem 8.2); Lehmer's conjectural universal floor is the smaller $\mu_L = 1.17628\dots$

5.2 The exact admissibility gate

Define the *degree* $\deg(m)$ and the integer *coefficient height* $\text{ch}(m) = \max_i |c_i|$. A *budget* is a pair $(D_{\max}, H_{\max}) \in \mathbb{Z}^2$.

Definition 5.3 (Northcott admissibility). m is *admissible* for $(D_{\max}, H_{\max})$ iff $\deg(m) \leq D_{\max}$ and $\text{ch}(m) \leq H_{\max}$.

Proposition 5.4 (Bounded capacity, exact gate). *The admissible set $\{m \in \mathbb{Z}[x] \text{ monic} : \deg m \leq D_{\max}, \text{ch}(m) \leq H_{\max}\}$ is finite, of size $\sum_{d=1}^{D_{\max}} (2H_{\max} + 1)^d$. The predicate of Definition 5.3 is decided entirely in $\mathbb{Z}$; the Mahler measure $M(m)$ never enters it, and the exact integer $\sum_i c_i^2$ certifies $M(m)^2 \leq \sum_i c_i^2$ by Theorem 5.2(b).*

Proof. Each of the d non-leading coefficients ranges over the $2H_{\max} + 1$ integers in $[-H_{\max}, H_{\max}]$, and the leading coefficient is 1; summing over $1 \leq d \leq D_{\max}$ gives the count. Finiteness is the constructive form of Northcott (Theorem 5.2(c)). The decision uses only the integer comparisons $\deg \leq D_{\max}$ and $\text{ch} \leq H_{\max}$. $\square$

Example 5.5 (Northcott finiteness, concretely). Under the budget $D_{\max} = 2$, $H_{\max} = 1$ the admissible monic integer polynomials number $\sum_{d=1}^2 3^d = 3 + 9 = 12$: the three linear $x + c$ ($c \in \{-1, 0, 1\}$) and the nine quadratics $x^2 + bx + c$ ($b, c \in \{-1, 0, 1\}$). Finiteness is not asymptotic—the entire admissible set is small and listable. Among the quadratics, $x^2 - x - 1$ (the golden seed φ) and $x^2 + x - 1$ are irreducible; $x^2 - 1$ and $x^2 + x$ are reducible and define no new field; $x^2 + 1$ cuts out $\mathbb{Q}(i)$ (Example 2.17). Raising $H_{\max}$ to 24 first admits $2\sqrt{6}$ ($x^2 - 24$); the budget is precisely the dial that bounds an explicitly enumerable dictionary.

Example 5.6 (Exact heights). The following are computed exactly (no roots, no floats):

seed	minimal polynomial	deg	ch	$\sum c_i^2$
φ	$x^2 - x - 1$	2	1	3
$2\sqrt{6}$	$x^2 - 24$	2	24	577
$\sqrt{7}$	$x^2 - 7$	2	7	50
$\text{gap} = \varphi^4$	$x^2 - 7x + 1$	2	7	51

Landau is tight here: $M(2\sqrt{6}) = 24$ and $24^2 = 576 \leq 577 = \sum c_i^2$.

Remark 5.7 (The float never crosses the gate). The Mahler measure is a float (it requires the complex roots). For instance the exact value $M(\sqrt{7}) = 7$ is computed by a root finder as 7.000000000000001—strictly above 7. A naive floating test “ $M \leq 7$ ” would erroneously reject $\sqrt{7}$ on rounding noise; the integer gate of Definition 5.3 cannot. The admissibility decision depends only on (deg, ch) and is invariant to the float Mahler value.

The growth gate `CAPACITY_DECISION` returns one of GROW/STOP/REJECT: REJECT if the seed is inadmissible (Definition 5.3); STOP if the residual defect is below a floor; and GROW otherwise. The floor is the subject of §8; in the shipped gate it is 0 (any nonzero defect passes), a choice we revisit there.

6 A worked learning episode

We trace the full loop in $K = \mathbb{Q}(\sqrt{2} + \sqrt{3}) = \mathbb{Q}[x]/(x^4 - 10x^2 + 1)$ (degree $n = 4$, so $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$), with forced basis $B = \{1, \theta\}$, $\theta = \sqrt{2} + \sqrt{3}$, so $\text{col}(B) = \mathbb{Q} \oplus \mathbb{Q}\theta$.

In-distribution. Any observation $a + b\theta$ ($a, b \in \mathbb{Q}$) lies in $\text{col}(B)$, so $\mathbf{r} = 0$: captured, nothing proposed.

Persistent off-axis novelty. Stream observations $\mathbf{x} = u + 2\sqrt{6}$ with $u \in \text{col}(B)$, where $2\sqrt{6} = \theta^2 - 5$ has coordinate vector $(-5, 0, 1, 0)$ and is G -orthogonal to $\text{col}(B)$. Then $\mathbf{r}(\mathbf{x}) = 2\sqrt{6}$ exactly, and, the trace being basis-independent,

$$\|\mathbf{r}\|_G^2 = T(2\sqrt{6}, 2\sqrt{6}) = \text{Tr}_{K/\mathbb{Q}}((2\sqrt{6})^2) = \text{Tr}_{K/\mathbb{Q}}(24) = 4 \cdot 24 = 96.$$

As an exact check, for $\mathbf{x} = \theta + 2\sqrt{6}$ the Pythagorean split (9) reads $\|\mathbf{x}\|_G^2 = 116 = 20 + 96$, with captured part $\|\theta\|_G^2 = \text{Tr}(\theta^2) = 20$, residual $\|\mathbf{r}\|_G^2 = 96$, and vanishing cross term $\text{Tr}(\theta \cdot 2\sqrt{6}) = 0$. Once the persistence count is met, *propose* returns the exact seed identity (Theorem 4.2) $m_{2\sqrt{6}} = x^2 - 24$, of degree 2 and coefficient height 24 (Example 5.6).

Capacity and threshold. Under a budget $(D_{\max}, H_{\max}) = (64, 256)$ the seed is admissible ($2 \leq 64$, $24 \leq 256$; Definition 5.3). With the information rule (14) and $\lambda = 2$, the gain 96 exceeds the cost $2 \log 24 = 6.36$ and the floor 0.562, so the decision is GROW.

Confirm. Adjoining $2\sqrt{6}$ to B and re-deriving P —the sole mutation (Protocol 4.1)—returns the residual of $u + 2\sqrt{6}$ to 0; the growth is appended to the witness chain.

Out-of-field novelty. Once the basis spans all of K , so the captured field is $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ itself with $G_K = \text{diag}(4, 8, 12, 24)$, suppose a quantity forced into $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{7})$ with a $\sqrt{7}$ component appears. Its field residual against K is nonzero, and the candidate generator $\sqrt{7}$ ($m = x^2 - 7$) is linearly disjoint from K (Example 4.7). Confirming the extension grows the field to the compositum with $G_{KL} = G_K \otimes G_L = \text{diag}(8, 56, 16, 112, 24, 168, 48, 336)$ (Proposition 4.5); the $\sqrt{7}$ residual then returns to 0, and the same height gate governs the decision on the actual compositum degree $4 \rightarrow 8$.

7 Information geometry of the trace form

We read G , and the residual norm, as Fisher information. The two structural facts are *classical*: $G = M^\top M$ (the trace form is the Minkowski–embedding Gram, Theorem 2.13; textbook algebraic number theory [9, 10]), and the Fisher information of a Gaussian location family $\mathcal{N}(Ma, I)$ is $M^\top M$ (the standard location–family formula [12]). Our part is to assemble these into a Fisher reading of the trace form, to pin the model–dependent conformal constant, and to draw the residual–norm–as–information–distance consequence for the learner—not to resolve an open mathematical problem. All displayed quantities are verified by exact symbolic computation (Appendix A); the choice of statistical model is not unique, and we are explicit about which constant each model produces.

7.1 Model 1: a Gaussian location family

Let the field coordinate $a \in \mathbb{R}^d$ parameterise the *mean* $\mu(a) = Ma$ of an isotropic Gaussian on the Minkowski space, $\mathbf{x} \sim \mathcal{N}(Ma, I_n)$, where M is the embedding matrix (5).

Theorem 7.1 (Trace form is Gaussian Fisher information). *For a totally real field, the Fisher information of the location family $\{\mathcal{N}(Ma, I_n)\}$ in the parameter a is*

$$\mathcal{I}_{\text{Gauss}} = M^\top M = G.$$

With covariance cI_n in place of I_n , $\mathcal{I}_{\text{Gauss}} = c^{-1}G$ (conformal, constant c).

Proof. For a Gaussian location family $\mathcal{N}(\mu(a), \Sigma)$, the Fisher information is $(\partial\mu/\partial a)^\top \Sigma^{-1} (\partial\mu/\partial a)$. Here $\partial\mu/\partial a = M$ and $\Sigma = I_n$, so $\mathcal{I} = M^\top M = G$ by Theorem 2.13; replacing Σ by cI_n scales by c^{-1} . $\square$

Proposition 7.2 (The discriminant is an information volume). *Under Model 1 the Fisher information is the constant metric G (Theorem 7.1), so its Jeffreys volume element—the reparameterisation–invariant prior density $\sqrt{\det \mathcal{I}}$ [16]—is the constant $\sqrt{\det G}$. The Jeffreys volume of one fundamental cell of the coordinate lattice (i.e. of $\mathcal{O}_K$ in an integral basis) is therefore*

$$\text{vol}_{\text{Jeffreys}} = \int_{[0,1]^n} \sqrt{\det G} \, da = \sqrt{\det G} = \sqrt{|d_K|}, \quad (11)$$

equal to the Minkowski covolume of $\sigma(\mathcal{O}_K)$ (§2.6); for a sub-order of index f it scales by f . The square root of the field discriminant is the natural Fisher–Jeffreys volume of one lattice cell of the statistical manifold.

Proof. For the location family $\mathcal{N}(Ma, I_n)$ the Fisher information is the constant $M^\top M = G$ (Theorem 7.1), independent of a ; the Jeffreys density is then $\sqrt{\det \mathcal{I}} = \sqrt{\det G}$ [16], and integrating this constant over a unit coordinate cell gives $\sqrt{\det G}$. By Theorem 2.13, $\det G = d_K$ for an integral basis (and $f^2 d_K$ for a sub-order of index f), and $\sqrt{\det G}$ is the covolume. $\square$

Remark 7.3 (Discriminant as “amount of arithmetic”). The discriminant measures the arithmetic complexity of K —its ramification. Equation (11) reads it as an information volume: the larger $|d_K|$, the more Fisher–Jeffreys volume one integer cell occupies, so a more ramified field is, in a precise statistical sense, “larger” to learn. The covolumes $\sqrt{|d_K|}$ tabulated in Table 2 ($\sqrt{5}$, $2\sqrt{2}$, $6\sqrt{3}$ for $\mathbb{Q}(\sqrt{5})$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt[3]{2})$) are exactly these information volumes.

Example 7.4 (Information volumes). By Proposition 7.2 the Jeffreys volume of one integer cell is $\sqrt{|d_K|}$. From Table 2: $\mathbb{Q}(\sqrt{5})$ has volume $\sqrt{5} \approx 2.236$, $\mathbb{Q}(\sqrt{2})$ has $2\sqrt{2} \approx 2.828$, and the more ramified $\mathbb{Q}(\sqrt[3]{2})$ has $6\sqrt{3} \approx 10.39$ —a single $\mathcal{O}_K$ -cell of $\mathbb{Q}(\sqrt[3]{2})$ thus occupies about 4.6 times the Fisher–Jeffreys volume of a cell of $\mathbb{Q}(\sqrt{5})$. “More ramified” is here literally “more information volume per lattice cell.”

Remark 7.5 (The field’s own description length). Proposition 7.2 prices the *field*, not just a seed. In an MDL/BIC account [14, 15] a model’s code length carries a term $\frac{1}{2} \log \det(\text{Fisher})$; here that is $\frac{1}{2} \log \det G = \frac{1}{2} \log |d_K| = \log \sqrt{|d_K|} = \log \text{covol}(\Lambda_K)$. The description length of working in K is thus the logarithm of its Minkowski covolume—an Occam penalty on ramification, complementing the per-seed entropy $\log M$ of §8.

Remark 7.6 (Information volume is the Model-1 reading). The volume $\sqrt{|d_K|}$ of Proposition 7.2 is the *Model-1* ($c = 1$) reading. Under the maximum-entropy Model 2 the metric on the residual subspace is $\mathcal{I}_{\text{exp}} = G/n$ (Theorem 7.9), whose volume element rescales one cell by $\det(G/n)^{1/2} = n^{-d/2} \sqrt{\det G}$. So “information volume = $\sqrt{|d_K|}$ ” is model-dependent in exactly the way Conjecture 7.13 describes: the discriminant reading belongs to the Gaussian model, the $n^{-d/2}$ -scaled reading to the conjugate model. Model 1’s manifold is moreover *flat* (constant Fisher G), so its content is the metric itself; any curvature of the statistical manifold lives in Model 2.

7.2 Model 2: the maximum-entropy family on the conjugates

The framework’s canonical, zero-parameter measure is *uniform* on the n Galois conjugates. (For seeds of growing degree this uniform-on-conjugates measure is the natural object of Bilu’s equidistribution theorem [18], under which the conjugates equidistribute on the unit circle and the Mahler measure is an integral over the limiting measure; it is natural here but not the only possible choice.) Consider the discrete exponential family $p(k; a) \propto \exp((Ma)_k)$ on outcomes $k \in \{1, \dots, n\}$, with sufficient statistic the embedding rows $s(k) = (\sigma_k(\omega_1), \dots, \sigma_k(\omega_d))$, and let $t = (\text{Tr}(\omega_1), \dots, \text{Tr}(\omega_d))^\top$ be the trace vector.

Theorem 7.7 (Conjugate-family Fisher information). *At the uniform point $a = 0$ (so $p \equiv 1/n$), the Fisher information of $\{p(\cdot; a)\}$ is*

$$\mathcal{I}_{\text{exp}} = \frac{1}{n} G - \frac{1}{n^2} t t^\top = \frac{1}{n} \left(G - \frac{1}{n} t t^\top \right). \quad (12)$$

Proof. For an exponential family the Fisher information equals the covariance of the sufficient statistic. At $a = 0$, p is uniform, so $\text{Cov}[s_i, s_j] = \frac{1}{n} \sum_k \sigma_k(\omega_i) \sigma_k(\omega_j) - (\frac{1}{n} \sum_k \sigma_k(\omega_i)) (\frac{1}{n} \sum_k \sigma_k(\omega_j)) = \frac{1}{n} G_{ij} - \frac{1}{n^2} t_i t_j$, using $\sum_k \sigma_k(\omega_i) \sigma_k(\omega_j) = G_{ij}$ (Theorem 2.13) and $\sum_k \sigma_k(\omega_i) = \text{Tr}(\omega_i) = t_i$. $\square$

The correction $\frac{1}{n^2} t t^\top$ is rank one, along the trace vector t . We isolate where it vanishes.

Lemma 7.8 (Trace-zero is the residual subspace). *For a coordinate vector v , $\langle v, 1 \rangle_G = t^\top v = \text{Tr}(v)$, where $1 = \omega_1$ is the constant and $\text{Tr}(v)$ denotes the trace of the field element with coordinates v . Consequently $v \perp_G 1 \iff \text{Tr}(v) = 0 \iff t^\top v = 0$. If the forced basis contains the constant ($1 \in \text{col}(B)$), then every residual $\mathbf{r} = \mathbf{x} - P\mathbf{x}$ satisfies $\text{Tr}(\mathbf{r}) = 0$.*

Proof. $\langle v, 1 \rangle_G = v^\top G e_1$. Since $\omega_1 = 1$, the first column of G is $(\text{Tr}(\omega_i \cdot 1))_i = (\text{Tr}(\omega_i))_i = t$, so $\langle v, 1 \rangle_G = t^\top v = \sum_i v_i \text{Tr}(\omega_i) = \text{Tr}(\sum_i v_i \omega_i) = \text{Tr}(v)$. If $1 \in \text{col}(B)$ then $\mathbf{r} \perp_G \text{col}(B) \ni 1$ (Proposition 3.2), so $\text{Tr}(\mathbf{r}) = 0$. $\square$

Theorem 7.9 (The residual norm is degree-scaled Fisher information). *On the trace-zero subspace $\{v : t^\top v = 0\}$, $G = n \mathcal{I}_{\text{exp}}$. In particular, if $1 \in \text{col}(B)$ then for every residual $\mathbf{r}$,*

$$\|\mathbf{r}\|_G^2 = n \cdot \mathcal{I}_{\text{exp}}(\mathbf{r}). \quad (13)$$

Proof. For $t^\top v = 0$, (12) gives $v^\top \mathcal{I}_{\text{exp}} v = \frac{1}{n} v^\top G v - \frac{1}{n^2} (t^\top v)^2 = \frac{1}{n} v^\top G v$, i.e. $v^\top G v = n v^\top \mathcal{I}_{\text{exp}} v$. By Lemma 7.8, when $1 \in \text{col}(B)$ the residual $\mathbf{r}$ is trace-zero, and (13) is this identity at $v = \mathbf{r}$. $\square$

7.3 Exact evidence

The matrices below are produced symbolically and verified exactly.

field (basis)	n	G	$\mathcal{I}_{\text{exp}}$
$\mathbb{Q}(\sqrt{5}), \{1, \varphi\}$	2	$\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$	$\begin{pmatrix} 0 & 0 \\ 0 & 5/4 \end{pmatrix}$
$\mathbb{Q}(\sqrt{2}, \sqrt{3}), \{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$	4	$\text{diag}(4, 8, 12, 24)$	$\text{diag}(0, 2, 3, 6)$
$\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{7})$	8	$\text{diag}(8, 56, 16, 112, 24, 168, 48, 336)$	$\text{diag}(0, 7, 2, 14, 3, 21, 6, 42)$

In each case $G = M^\top M$, $\mathcal{I}_{\text{exp}} = \frac{1}{n} G - \frac{1}{n^2} t t^\top$, and $G = n \mathcal{I}_{\text{exp}}$ on the trace-zero subspace. For a concrete residual, $\sqrt{5} = 2\varphi - 1$ has coordinates $(-1, 2)$, $\text{Tr}(\sqrt{5}) = 0$, and

$$\|\sqrt{5}\|_G^2 = 10 = 2 \cdot 5 = n \cdot \mathcal{I}_{\text{exp}}(\sqrt{5}),$$

matching (13) with $n = 2$.

7.4 Relative entropy and natural gradient

Both identifications are local forms of relative entropy. For a smooth family $\{p_a\}$ the Kullback–Leibler divergence expands to second order as

$$D_{\text{KL}}(p_a \parallel p_{a+\delta}) = \frac{1}{2} \delta^\top \mathcal{I}(a) \delta + O(\|\delta\|^3),$$

so the Fisher matrix is the Riemannian metric of the statistical manifold [13]. By Theorems 7.1 and 7.9, G (respectively $n^{-1}G$ on the residual subspace) is that metric. Consequently a step measured in $\|\cdot\|_G$ is a natural-gradient step [12], and $\|\mathbf{r}\|_G^2$ is twice the leading Kullback–Leibler cost of failing to capture $\mathbf{r}$. This is the precise sense in which the residual norm is an information distance rather than a merely Euclidean one; it is the analytic content used in §8.

Proposition 7.10 (The projector is a one-step natural-gradient fit). *Fix $x \in K$ and the forced basis B , and consider the quadratic loss $L(a) = \frac{1}{2}\|x - Ba\|_G^2$ on coordinate vectors a . In the G -geometry its gradient and Hessian are $\nabla L(a) = B^\top G(Ba - x)$ and $\nabla^2 L = B^\top GB$, so the natural-gradient (Newton) step from $a = 0$ lands at $a^* = (B^\top GB)^{-1}B^\top Gx$. Then $Ba^* = Px$ and the optimal residual is $x - Px = \mathbf{r}$: the trace-form natural gradient solves the exact-fit problem in a single step, and its fixed point is the projector of Proposition 3.2.*

Proof. The gradient and Hessian are immediate, and L is quadratic, so the single Newton step $a^* = -(\nabla^2 L)^{-1}\nabla L(0) = (B^\top GB)^{-1}B^\top Gx$ is exact; then $Ba^* = B(B^\top GB)^{-1}B^\top Gx = Px$ by the definition of P , and the residual is $x - Px = \mathbf{r}$. $\square$

Remark 7.11. Because the loss is quadratic with Hessian G , the natural gradient *is* the Newton step: learning the captured part of x takes one exact step, not an iterative descent. This is the optimisation shadow of the idempotence $P^2 = P$ (Proposition 3.2)—re-fitting an already-captured x moves nothing.

Example 7.12 (One-step fit in $\mathbb{Q}(\sqrt{5})$). Take $B = \langle \varphi \rangle$ (so $B = (0, 1)^\top$), $G = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$, and target $x = \sqrt{5} = (-1, 2)^\top$ (Example 3.3). The natural-gradient step is $a^* = (B^\top GB)^{-1}B^\top Gx = 3^{-1} \cdot 5 = \frac{5}{3}$, giving $Px = (0, \frac{5}{3})^\top$ and residual $\mathbf{r} = (-1, \frac{1}{3})^\top$ with $\|\mathbf{r}\|_G^2 = \frac{5}{3}$ in a single move—exactly the projector of Example 3.3, now realised as a one-step Newton fit. A Euclidean step would use $B^\top B = 1$ in place of $B^\top GB = 3$ and land off the G -projection.

7.5 Honest caveats

Conjecture 7.13 (No canonical model). There is no model-independent identification of G with “the” Fisher metric of the substrate. Two natural models give two constants: $G = \mathcal{I}_{\text{Gauss}}$ ($c = 1$, whole space, Theorem 7.1) and $G = n\mathcal{I}_{\text{exp}}$ ($c = n$, residual subspace, Theorem 7.9). Which model is canonical is an interpretive choice, not forced by the field. By Čencov’s theorem the Fisher metric is the unique Riemannian metric invariant under sufficient statistics *up to scale* [19]; the $c = 1$ versus $c = n$ ambiguity is therefore a choice of model *and* of that scale, not a choice of a different metric. The companion paper names a third, framework-native scale—the self-action eigenframe gap $c = \sqrt{1 + 4C}/(2C)$ at a critical point of the ladder $x^2 + x = C$ (golden gate $c = \sqrt{5}/2$)—under which the exchange rate is the identity $\lambda = 2c$; by the present conjecture this is a principled *selection* of c , not a removal of the freedom it identifies.

Remark 7.14 (Two further caveats). (i) *Complex embeddings.* Theorems 2.13–7.9 are stated for totally real fields. For $r_2 > 0$ the trace form is indefinite; the positive-definite Hermitian form $G_2 = M^*M$ of §2.5 replaces G , and the identities carry over with G_2 in its place—the isotropic Gaussian of Theorem 7.1 taken on the canonical real Minkowski space $\mathbb{R}^n$, whose embedding Jacobian J satisfies $J^\top J = G_2$. The worked examples in this section are totally real and use G directly; an exact probe on a field with $r_2 > 0$ is left to future work (O2). (ii) *The $1 \in \text{col}(B)$ condition.* Identity (13) requires the constant to lie in the forced basis (Lemma 7.8). The constant is *not* auto-injected; “ $\mathbf{r} \perp_G 1$ ” is therefore a usage convention, true only when $1 \in \text{col}(B)$. When $1 \notin \text{col}(B)$ the residual can carry a nonzero-trace component and (13) fails. This is verified both ways in the probe: in $\mathbb{Q}(\sqrt{5})$, projecting onto $\langle 1 \rangle$ gives a trace-zero residual, while projecting onto $\langle \varphi \rangle$ leaves the residual of 1 with trace $5/3 \neq 0$.

8 The growth threshold

We now ask the operational question (the “residual–enlargement” rule, denoted R1 in the project): how large must $\|\mathbf{r}\|_G$ be to justify adjoining a seed of arithmetic height? We frame it as information gained versus information spent, using Theorem 7.9.

8.1 Gain versus cost

Definition 8.1 (Information gain and cost). For a residual $\mathbf{r}$ (with $1 \in \text{col}(B)$) and a candidate seed θ of degree d and Mahler measure $M(\theta) = H(\theta)^d$, set

$$\text{gain} = \|\mathbf{r}\|_G^2 = n \mathcal{I}_{\text{exp}}(\mathbf{r}), \quad \text{cost} = \lambda \log M(\theta),$$

where $\log M(\theta)$ is the seed’s topological entropy [7] (its intrinsic description length) and $\lambda > 0$ is an information exchange rate.

Theorem 8.2 (A provable nonzero cost floor). *For every non–reciprocal monic $p \in \mathbb{Z}[x]$ that is not cyclotomic, $M(p) \geq \mu_S = 1.3247179\dots$, the real root of $x^3 - x - 1$ (the plastic number). Hence the cost of any non–reciprocal genuine seed is at least $\lambda \log \mu_S > 0$.*

Proof. This is Smyth’s theorem [3]; μ_S is the smallest Mahler measure of a non–reciprocal algebraic integer and is the real root of $x^3 - x - 1$. Positivity of $\log \mu_S$ is immediate from $\mu_S > 1$. $\square$

Remark 8.3 (The reciprocal case is the unsolved core). Smyth’s floor is unconditional *only for non–reciprocal* integers. The hard, unsolved part of Lehmer’s problem is exactly the reciprocal case: every known Mahler measure in $(1, \mu_S)$ comes from a *reciprocal* integer, and Salem numbers and units—which arise naturally as seeds—are reciprocal. For reciprocal seeds there is *no proven uniform positive floor*; one has only Dobrowolski’s degree–dependent bound (Remark 8.5), which tends to 1 as the degree grows and so is not uniform, or Lehmer’s conjectural μ_L (Conjecture 8.4). Hence “the cheapest genuine seed costs at least $\lambda \log \mu_S$ ” is theorem–backed only for non–reciprocal seeds. The worked seeds of this paper— $\varphi(x^2 - x - 1)$, $2\sqrt{6}(x^2 - 24)$, $\sqrt{7}(x^2 - 7)$ —are all non–reciprocal (their root sets are not closed under $\alpha \mapsto \alpha^{-1}$), so their floors are unconditional; only the *general* positive–floor claim carries the caveat.

Conjecture 8.4 (Lehmer). $\inf\{M(p) : M(p) > 1, p \in \mathbb{Z}[x] \text{ monic}\} = \mu_L = 1.17628081\dots$, the largest real root of Lehmer’s degree–10 polynomial $x^{10} + x^9 - x^7 - x^6 - x^5 - x^4 - x^3 + x + 1$. This would extend Theorem 8.2 to all (including reciprocal) seeds; it is *open*.

Remark 8.5 (Dobrowolski). Unconditionally, $M(p) \geq 1 + c(\log \log d / \log d)^3$ for non–cyclotomic p of degree d [4]; this is a degree–dependent (vanishing) lower bound, weaker than a uniform floor but provable for all degrees.

Remark 8.6 (Lehmer as a spectral gap). Reading $\log M$ as an energy, $\log M(\theta) = \sum_{|\alpha_i| > 1} \log |\alpha_i|$ is the topological entropy of the companion toral automorphism [7]. Lehmer’s Conjecture 8.4 then asserts a *spectral gap* $(0, \log \mu_L)$ in the energy spectrum of monic integer objects: no non–trivial seed carries entropy below $\log \mu_L$. The growth floor of (14) is exactly this gap transported to the gain side; Smyth’s Theorem 8.2 makes a (non–reciprocal) gap unconditional.

A minimum–description–length reading. The criterion $\text{gain} \geq \text{cost}$ is an MDL test [14]. Capturing $\mathbf{r}$ shortens the description of the data by its explained information $\|\mathbf{r}\|_G^2$ (twice the Kullback–Leibler cost, §7), while *naming* the new seed costs its own description length—its entropy $\log M(\theta)$. One grows precisely when the code saved exceeds the code spent. The exchange rate λ converts between these two ledgers; the Lehmer/Smyth floor makes the seed’s code length bounded below, an arithmetic Occam penalty.

8.2 The derived rule

Combining the exact admissibility gate (Proposition 5.4) with the information criterion:

REJECT if θ is inadmissible (degree or coefficient height over budget); STOP if $\ \mathbf{r}\ _G^2 < \lambda \log M(\theta)$ (in particular if $< \lambda \log \mu_S$, below the cheapest seed); GROW if admissible and $\ \mathbf{r}\ _G^2 \geq \lambda \log M(\theta)$.	(14)
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This *refines* the shipped gate, which uses the floor 0 and never stops a nonzero residual, with a nonzero, seed-dependent threshold.

8.3 Evidence and subsumption

Take the representative rate $\lambda = 2$ (an MDL/BIC-scale value, conjectural; see Remark 8.11). With $\log \mu_S = 0.2812\dots$ the constant floor is $\lambda \log \mu_S = 0.562\dots$

residual / seed	n	gain = $\ \mathbf{r}\ _G^2$	M	cost = $\lambda \log M$	decision
$2\sqrt{6}$ (shipped GROW)	4	96	24	6.36	GROW
$\sqrt{7}$ (shipped GROW)	8	56	7	3.89	GROW
tiny off-axis	4	1/10	—	—	STOP (0.1 < 0.562)

The shipped growth cases clear the threshold by a wide margin, so (14) *subsumes* the existing behaviour. The tiny-residual row is where the principled floor differs from 0: the current gate would grow on it, the refined gate stops.

Example 8.7 (Lattice-aligned noise). The tiny-residual row is not hypothetical. In $\mathbb{Q}(\sqrt{5})$ take the residual centroid $\mathbf{r} = \frac{1}{10}\sqrt{5}$, an exact rational multiple of the off-axis lattice direction $\sqrt{5} = 2\varphi - 1$. It is trace-zero ($\text{Tr}(\mathbf{r}) = \frac{1}{10}\text{Tr}(\sqrt{5}) = 0$, so the Fisher reading of §7 applies) and *exactly* off-axis, yet arbitrarily small: $\|\mathbf{r}\|_G^2 = \frac{1}{100}\|\sqrt{5}\|_G^2 = \frac{1}{100} \cdot 10 = \frac{1}{10}$. The shipped gate (floor 0) would GROW on it—noise that merely happens to land on the lattice—while the Smyth floor $\lambda \log \mu_S = 0.562$ STOPS it, since $\frac{1}{10} < 0.562$. This is the concrete motivation for a nonzero floor: it rejects sub-threshold lattice-aligned noise that floor = 0 cannot (Figure 4; probe, §A).

8.4 A degree-aware floor

Remark 8.8 (Normalising across fields). Because $\|\mathbf{r}\|_G^2 = n \mathcal{I}_{\text{exp}}(\mathbf{r})$ (Theorem 7.9), a *constant* floor on $\|\mathbf{r}\|_G^2$ corresponds to a Fisher-significance threshold of order $1/n$: in a high-degree field a residual that is intrinsically tiny still clears a constant floor, because the factor n inflates the norm. A *degree-aware* floor $n \lambda \log \mu_S$ keeps the Fisher-significance threshold constant across fields. With $\lambda = 2$ the degree-aware floor is 2.25 at $n = 4$ and 4.50 at $n = 8$. A gain of 1 clears the constant floor 0.562 but not the degree-aware floor 4.50 at $n = 8$. The shipped cases (gain = 96, 56) clear either floor; the degree-aware refinement is for robustness, not for the present outcomes.

8.5 The exact-versus-float boundary, sharpened

Remark 8.9 (The threshold is intrinsically float). The admissibility half of (14) is exact (Proposition 5.4). The information half is not: $\log M(\theta)$ is transcendental, so $\|\mathbf{r}\|_G^2 \geq \lambda \log M$ cannot be a pure $\mathbb{Q}/\mathbb{Z}$ decision. The gain $\|\mathbf{r}\|_G^2$ remains an exact rational; only the cost is real. An exact

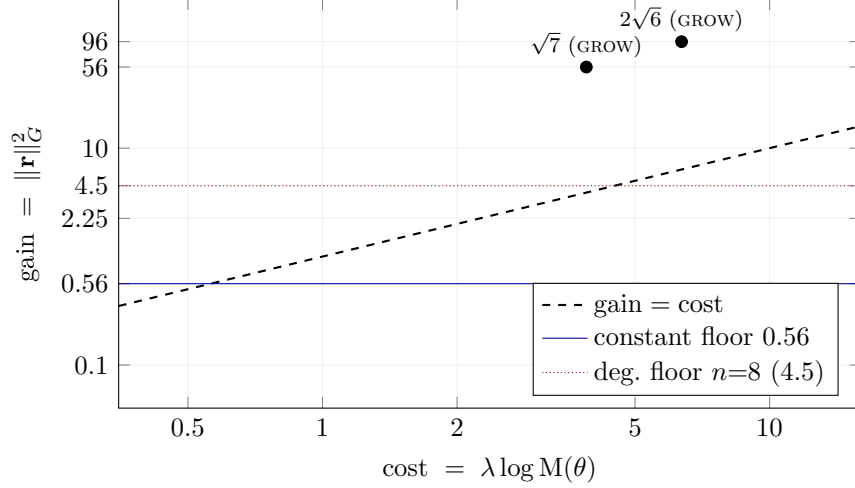


Figure 4: The growth decision in the (cost, gain) plane (log–log). The shipped cases $2\sqrt{6}$ (96 vs 6.356) and $\sqrt{7}$ (56 vs 3.892) sit far above both the boundary gain = cost and the floors, hence GROW; the lattice-aligned tiny residual (gain = 0.1) sits below the constant floor 0.562 and is STOPPED (Example 8.7). The degree-aware floor is 2.25 at $n=4$ and 4.5 at $n=8$ (Remark 8.8). All gain/cost values for the grow cases and the floors are probe-verified; the tiny-residual mark’s horizontal position is illustrative (its decision is the gain-versus-floor comparison, independent of cost).

gate would therefore require *certified interval arithmetic* on $\log M$ (so that the comparison becomes a certified rational decision), which we carry out on a representative case below (Example 8.10). Thus capacity is exact while the information threshold is, by its nature, analytic—but still *decidable* with rigor.

Example 8.10 (A certified GROW decision). We perform the certified comparison for $\sqrt{7}$ (seed $x^2 - 7$, $M = 7$) at $\lambda = 2$. Interval arithmetic yields the rigorous rational enclosure

$$\log 7 \in [1.94591, 1.94592], \quad \text{whence} \quad \lambda \log 7 \leq 2 \cdot 1.94592 = 3.89184.$$

The gain $\|\mathbf{r}\|_G^2 = 56$ is an exact rational, and $56 \geq 3.89184 \geq \lambda \log 7$, so GROW is *certified* with no trust in any rounded value: an exact rational dominates a rigorous *upper* bound on the cost. The same procedure certifies $2\sqrt{6}$ (seed $x^2 - 24$): $\log 24 \in [3.17805, 3.17806]$ gives $\lambda \log 24 \leq 6.35612 \leq 96$. Both enclosures are produced and checked by the probe (§A); a sharper enclosure is needed only when gain and cost fall within one interval width of each other.

Remark 8.11 (The exchange rate is conjectural). The rate λ is the information price of one parameter. The minimum-description-length principle [14] and the Bayesian information criterion [15] suggest scales of order 1 to $\frac{1}{2} \log N$, but the value appropriate to this substrate is not derived here. Likewise the effective floor constant—Lehmer’s μ_L (conjectural), Smyth’s μ_S (provable, non-reciprocal), or Dobrowolski’s degree-dependent bound—is a modelling choice. These, together with Conjecture 7.13, are the locus of what remains genuinely open in R1.

9 Applications and use cases

9.1 Streaming anomaly and novelty detection

The residual norm is a direct novelty score: $\|\mathbf{r}(\mathbf{x})\|_G^2 = 0$ for in-distribution (captured) observations and is strictly positive for off-axis novelty. A calibration-gated detector emits an alert only when (a) the score exceeds threshold and (b) a calibration predicate passes, and surfaces persistent novelty as a growth *proposal* via Protocol 4.1 without ever auto-acting. The detector is exact (a float observation is rejected at intake), and it inherits the witness chain: every surfaced proposal is recorded. By Theorem 7.9 the score is, when $1 \in \text{col}(B)$, a degree-scaled Fisher information distance—so “how anomalous” has an information-geometric meaning, not merely a Euclidean one.

Example 9.1 (A calibration-gated detection). Take the learner of §6 as the in-distribution model: observations in $\text{col}(B) = \mathbb{Q} \oplus \mathbb{Q}\theta$ score $\|\mathbf{r}\|_G^2 = 0$ and raise no alert. Inject the off-axis $2\sqrt{6}$ (score 96). While the calibration predicate is closed, the detector records the novelty but *suppresses* the alert; once calibration passes, the same observation raises an alert and surfaces the growth proposal $x^2 - 24$ —never auto-confirmed. The basis is unchanged until an explicit confirm, and the surfaced proposal is appended to the witness chain. Thus “anomalous” is the exact, calibration-gated event $\|\mathbf{r}\|_G^2 > 0$, whose magnitude is an information distance (Theorem 7.9).

Example 9.2 (A streaming trace). Continuing in $K = \mathbb{Q}(\theta)$, $\theta = \sqrt{2} + \sqrt{3}$, with forced basis $B = \{1, \theta\}$ and a calibration predicate that opens after the third observation. Here $2\sqrt{6} = \theta^2 - 5 \in \text{col}(B)^{\perp_G}$ carries the off-axis score 96 (§6), and “ $1 + 2\sqrt{6}$ ” has the same residual as $2\sqrt{6}$ because its 1-part already lies in $\text{col}(B)$.

t	observation	$\ \mathbf{r}\ _G^2$	calibrated	action
1	$2 - \theta$ (in $\text{col } B$)	0	no	—
2	$3 + 4\theta$ (in $\text{col } B$)	0	no	—
3	$1 + 2\sqrt{6}$ (off-axis)	96	no	novelty recorded, alert <i>suppressed</i>
4	$1 + 2\sqrt{6}$	96	yes	alert; <i>propose</i> seed $x^2 - 24$
5	confirm	—	yes	basis grows to $\{1, \theta, 2\sqrt{6}\}$; witnessed
6	$1 + 2\sqrt{6}$	0	yes	captured

The score is an exact rational at every step; the basis is mutated only at $t = 5$, and only by an explicit confirm. Steps 3 and 4 show the calibration gate decoupling *detection* (score > 0) from *alerting*, and step 6 shows capture collapsing the score to 0 once the seed is adjoined.

9.2 A capacity-bounded dictionary learner

Protocol 4.1 with the gate of §5 is an online dictionary learner whose capacity is bounded *by arithmetic*, not by a hyperparameter: by Proposition 5.4 only finitely many seeds are admissible under a fixed budget, so the model cannot silently inflate. Persistence guards against noise (a residual that averages to zero proposes nothing); the height gate guards against complexity; the compositum mechanism (§4.4) extends the same discipline across fields.

9.3 Computational aspects

Every operation is exact rational (or integer) arithmetic, trading floating speed for reproducibility; we record the costs. Forming G from a basis is $O(n^3)$ rational multiplications, or $O(n^2)$ trace

evaluations given a multiplication table. The projector solve $P = B(B^\top GB)^{-1}B^\top G$ is an $O(n^3)$ exact linear solve; in streaming use one appends a column to B and updates the small factorisation of $B^\top GB$ rather than refactoring. The coordinates-to-minimal-polynomial bridge (Theorem 4.2) is a Smith normal form of the $n \times n$ matrix $xI - \rho(\alpha)$ over $\mathbb{Q}[x]$; with fraction-free (Bareiss-style) elimination the intermediate coefficients stay controlled and the cost is polynomial in n and the bit-size of α 's coordinates. The admissibility gate is $O(n)$ integer comparisons (Proposition 5.4). The only inexact step is the threshold comparison $\|\mathbf{r}\|_G^2 \geq \lambda \log M(\theta)$, evaluated by certified interval arithmetic on $\log M$ (§8) while the gain side stays an exact rational. The standing practical caveat is rational coefficient growth: exact Gram and projector entries can acquire large numerators and denominators, so an implementation bounds the working degree and height (which the capacity gate already enforces) or tests equalities modulo a prime and lifts. None of this alters a decision—it only prices exactness.

9.4 Natural gradient and model selection

Theorem 7.1 says the trace form is a Fisher metric; consequently moving in the G -geometry is a *natural-gradient* step in the sense of Amari [12, 13], not an ad hoc Euclidean one. The growth rule (14) is a model-selection criterion in the MDL/BIC family [14, 15]: adjoining a parameter (a basis direction) is justified exactly when the information it explains, $\|\mathbf{r}\|_G^2$, exceeds its description length $\lambda \log M(\theta)$. The Lehmer floor (Theorem 8.2, Conjecture 8.4) gives this description length a strictly positive minimum—an arithmetic Occam bound.

9.5 The trace form as an exact integer kernel

The trace pairing makes K a reproducing structure in the sense of kernel methods—but exactly and over $\mathbb{Z}$. The Minkowski embedding σ is a *feature map* $K \rightarrow \mathbb{R}^n$, and the trace form is its kernel,

$$T(x, y) = \text{Tr}(xy) = \langle \sigma(x), \sigma(y) \rangle = \sum_k \sigma_k(x) \sigma_k(y),$$

positive definite on totally real fields and integer-valued on $\mathcal{O}_K$. The forced subspace $\text{col } B$ is the span of the chosen “support vectors”; the projector P is the kernel projection onto them; and $\|\mathbf{r}\|_G^2$ is the squared kernel distance to that span. Three properties separate this from an analytic kernel: it is *exact* (decisions over $\mathbb{Q}$, not floating Gram inversions); it is *integer-valued* on the ring of integers, so its Gram is a lattice Gram with $\det = d_K$; and it *carries arithmetic*—the same kernel fixes the discriminant (Proposition 7.2), the different (§3.3), and the height floor (§8) that an inner product alone does not see. Learning is kernelised dictionary growth in which the kernel is a number field.

9.6 Persistence as exact re-derivation

A trained model is, in its entirety, a short list of monic-integer minimal polynomials—the forced anchors—together with the residual deltas accumulated since the last anchor. Because the geometry is exact, a restart is *re-derivation*, not deserialisation: from the anchor polynomials one recomputes the integral basis, the trace-form Gram G , and the projector P bit-for-bit (Theorems 2.13, 3.2), then replays the deltas. Two consequences are mathematical, not engineering. First, the reconstructed G and P are *identical* to the originals, so a model and its restart agree as operators, not merely to within a tolerance—there is no approximation to drift. Second, the stored state is a self-certificate: the witness chain of Protocol 4.1 makes the whole learning history replayable and

tamper-evident, since each growth step is reproducible from its seed by exact arithmetic. This is the precise sense in which the substrate is exact “all the way down”—the persisted object is not an opaque parameter blob but arithmetic that regenerates itself.

9.7 Toward a live, additive deployment

The learner is intended to be wired into a host service additively: read-only *observe* and *propose* endpoints, a single calibration-gated *confirm*, and a persistence model that stores a model as forced anchors (the seed minimal polynomials) plus residual deltas, so that restart is exact re-derivation rather than blob reload. We mention this to indicate the intended scope; it is engineering, not mathematics, and is out of scope for the present paper.

10 Open problems

- (O1) **The canonical model (Conjecture 7.13; partially answered in the companion).** Is there a principled reason to prefer $c = 1$ (Gaussian) or $c = n$ (max-entropy on conjugates)? The companion paper supplies a third, framework-native reading—the scale of the self-action eigenframe at a critical point of the ladder $x^2 + x = C$, namely $c = \sqrt{1 + 4C}/(2C)$ (golden gate $c = \sqrt{5}/2$)—and shows the exchange rate is then the identity $\lambda = 2c$. By Conjecture 7.13 (Čencov) this remains a principled *choice* of scale, not a derivation: it names a preferred c without removing the irreducible freedom the conjecture identifies.
- (O2) **Complex embeddings (partially resolved, §2.5).** The Hermitian form $G_2 = M^*M$ treats $r_2 > 0$ in principle, and the Fisher identities transfer to it (Remark 7.14). What remains is an exact, probe-level verification on a field with $r_2 > 0$ and confirmation that the max-entropy ($c = n$) model transfers cleanly when the conjugates are genuinely complex.
- (O3) **The exchange rate λ .** Derive (rather than posit) the information price of a basis direction for this substrate (Remark 8.11).
- (O4) **The non-disjoint compositum (Remark 4.8).** Construct the true compositum Gram when the generator’s minimal polynomial factors over K (via a primitive element, resultants, and factorisation over a number field).
- (O5) **Applying the degree-aware Lehmer floor.** Replace the shipped floor 0 by the information floor of §8, computing $\log M$ by certified interval arithmetic and conditioning the gate on $1 \in \text{col}(B)$ (Remark 8.8).

Remark 10.1 (Scope and limitations). We restate the boundary of what is claimed. The exact layer—membership, the residual, the projector, the coordinates-to-minimal-polynomial bridge, and admissibility—is decided over $\mathbb{Q}$ and $\mathbb{Z}$ and is proved or machine-verified. The *thresholding* layer is not exact: $\log M(\theta)$ is transcendental, the exchange rate λ is posited rather than derived, and—crucially—the cost floor is provably positive only for *non-reciprocal* seeds (Smyth); for reciprocal seeds, such as Salem numbers and units, no uniform positive floor is known (Remark 8.3; Lehmer’s conjecture). The information-geometry reading is conformal with a model-dependent constant ($c = 1$ or $c = n$), not a unique identity, and requires $1 \in \text{col}(B)$. The compositum construction is proved only in the linearly disjoint case. Finally, the framework yields *finiteness* of the admissible seed set (Northcott, Proposition 5.4) but no convergence-rate or sample-complexity guarantee—how quickly a field is learned from observations is not bounded here, and is left to future work. None of these gaps touches the exact core; each is flagged where it arises and collected in §10.

A Reproducibility

Every numerical claim above is reproduced by a machine-checked probe; the exact symbolic and rational computations are the evidence for the statements marked as verified.

- **Trace forms, embeddings, Fisher matrices** (Theorems 2.13, 7.1–7.9; §7 table): a symbolic probe builds the embedding matrices M , verifies $G = M^T M$, computes $\mathcal{I}_{\text{exp}} = \frac{1}{n}(G - \frac{1}{n}tt^T)$, and asserts $G = n\mathcal{I}_{\text{exp}}$ on the trace-zero subspace, including $\|\sqrt{5}\|_G^2 = 10$ and the $1 \in \text{col}(B)$ both-ways check (Remark 7.14).
- **Exact heights and admissibility** (Theorem 5.2, Proposition 5.4, Example 5.6): an integer-only probe checks $\deg$, ch , $\sum c_i^2$, the Landau certificate $M^2 \leq \sum c_i^2$ (cross-checked against the float Mahler), the rejection of float inputs, and the Northcott count.
- **The growth threshold** (§8, (14)): a probe checks that the shipped cases clear $\|\mathbf{r}\|_G^2 \geq \lambda \log M$, that the Smyth floor is a positive minimum, that a sub-floor residual is stopped by the refined gate but not by floor 0, and that the degree-aware floor differs from the constant floor (the values 0.562, 2.25, 4.50).
- **The coordinates-to-minimal-polynomial bridge** (Theorem 4.2, Example 4.3) and the compositum (Proposition 4.5, Example 4.7) are exercised by the learner’s own tests, including the $\sqrt{7}$ growth to $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{7})$ with $G = G_K \otimes G_L$.
- **Spectral, discriminant-volume, certified threshold, catalog, figures** (Propositions 2.3, 7.2; Remark 2.4; Examples 8.10, 8.7; Table 2; Figures 2, 3, 4; probe `test_paper_claims.py`, `sympy+mpmath`): asserts $M\rho(x)M^{-1} = \text{diag}(\sigma_k(x))$ and $\text{charpoly}(\rho(\theta)) = m_\theta$; $\det G = d_K$ (or $\text{index}^2 d_K$) and the Jeffreys volumes $\sqrt{|d_K|}$ for the catalog; $N(\mathfrak{d}) = |d_K|$; the spectral Mahler values $M(\sqrt{2} + \sqrt{3}) = 5 + 2\sqrt{6}$ and $M(\psi) = \mu_S$; the rigorous enclosures $\log 7 \in [1.94591, 1.94592]$, $\log 24 \in [3.17805, 3.17806]$ with the certified GROWS; and the lattice-aligned-noise decision $\frac{1}{10} < 0.562$.

Provenance. The information-geometry results (this paper’s §7) and the threshold derivation (§8) correspond to verification probes committed at `d526098` (Fisher) and `c5605d7` (threshold); the exact arithmetic core, the height/capacity gate, the bridge, and the compositum were committed earlier in the same line of work. The expansion’s new numbers—the figures, the spectral and discriminant-volume results, the certified-interval decisions, and the fields catalog—are reproduced by the additive probe `test_paper_claims.py` (`sympy` and `mpmath`). All numerical values displayed here match those probes.

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Statement	Status	Where
$G = M^\top M$ (trace form = embedding Gram)	Classical (cited) + probe	Thm 2.13
$\text{spec } \rho(x) = \{\sigma_k(x)\}$; M diagonalises ρ	Classical (cited) + probe	Prop 2.3
$\det G = d_K$ (integral basis; $[\mathcal{O}_K : \Lambda]^2 d_K$ for a sub-order); $\text{covol} = \sqrt{ d_K }$ (index $\cdot \sqrt{ d_K }$ for a sub-order)	Classical (cited) + probe	Thm 2.13, Tab 2
$N(\mathfrak{d}) = d_K $ (different)	Classical (cited) + probe	Tab 2
Discriminant = Jeffreys volume $\sqrt{\det G}$ (integral basis)	Synthesis (proved)	Prop 7.2
$G = \text{Fisher info of a Gaussian location family } (c = 1)$	Classical (cited) + probe	Thm 7.1
$\ \mathbf{r}\ _G^2 = n \mathcal{I}(\mathbf{r})$ on the residual subspace ($c = n$)	This work + probe	Thm 7.9
$M(\theta) = \prod_{ \lambda > 1} \lambda $ (spectral)	Classical (cited) + probe	Rem 2.4
$\log M = \text{toral topological entropy}$	Classical (cited)	Prop 2.5
Smyth floor $\mu_S = 1.3247$ (<i>non-reciprocal only</i>)	Cited (theorem)	Thm 8.2
Reciprocal–seed floor (Salem numbers, units)	Open (no uniform floor)	Rem 8.3
Certified GROW $(\sqrt{7}, 2\sqrt{6})$ via interval $\log M$	Verified (interval)	Ex 8.10
Compositum $G = G_K \otimes G_L$ (disjoint)	This work (proved)	Prop 4.5
Bridge $m_\alpha = \text{largest invariant factor of } xI - \rho(\alpha)$	This work (proved)	Thm 4.2
Canonical statistical model (c ; Čencov: free up to scale)	Conjectural; named $c \in \{1, n, \sqrt{1 + 4C}/(2C)\}$	Conj 7.13
Exchange rate $\lambda = 2c$ (identity; see companion)	Model-dependent through c only	Rem 8.11
Lehmer floor $\mu_L = 1.176$	Conjectural	Conj 8.4
Non-disjoint compositum Gram	Open	Rem 4.8
Fisher identities for $r_2 > 0$	Open (probe-level)	Rem 7.14

Table 3: Honesty ledger. *Classical (cited)*: a standard result we use and re-verify by probe, not a contribution of this paper. *This work/Synthesis*: assembled or proved here—the Fisher reading, the residual identity, the discriminant-as-volume framing, and the bridge/compositum constructions. *Verified (interval)*: a certified rational decision. *Cited (theorem)*: Smyth’s floor, unconditional only for *non-reciprocal* seeds. *Conjectural/model-dependent/Open*: the genuinely unsettled parts. Every “+ probe” row is reproduced by a machine-checked computation (§A).

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